Uruguay and the Fossilization Meme Testing Ultraactivity Beyond Argentina

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ABSTRACT

Why do democracies fossilize? Argentina attempts 50 structural reforms since 1989; only 4 succeed (8%). Uruguay attempts 54 reforms since 1985; 34 succeed (63%).
The difference: ultraactivity. Ultraactivity means legal norms persist indefinitely beyond their stated term without democratic renewal. Argentina embedded ultraactivity in labor law (1958); Uruguay eliminated it (1991).
Statistical analysis shows elimination increased Uruguay's reform success rate by 42 percentage points. But Uruguay still fails 37% of reforms.
Why? Eliminating the most toxic fossilization mechanism reveals others: mandatory referendums operate as asymmetric ratchets, allowing benefit expansion but blocking contraction.
The theoretical puzzle: non-retroactivity (criminal law cannot apply retroactively) diffused to 97% of constitutions in 261 years. Its logical twin—non-ultraactivity (norms cannot bind indefinitely without renewal)—appears in only 18%. Both protect against temporal tyranny. Both are equally fundamental to democratic legitimacy. Yet one succeeds universally; the other fails. Why? Memetic fitness depends on who benefits today, not normative superiority.
Present majorities protect themselves; future majorities cannot.

Keywords: Ultraactivity, legal memes, extended phenotype, constitutional lock-in, Uruguay, pension reform, referendum ratchet, memetic theory, veto points, institutional fossilization

JEL Codes: D72 (Political Processes); H55 (Social Security); K00 (General Law); K10 (Basic Areas of Law); P16 (Political Economy); Z13 (Economic Sociology)

I. INTRODUCTION

The Puzzle of Democratic Paralysis

Democracies are supposed to adapt. When policies fail, voters replace governments. New governments implement new policies. The system corrects itself. That's the theory.

Reality often disappoints. Some democracies get stuck. They identify problems correctly, propose reasonable solutions, pass legislation; and then watch as reforms collapse, reverse, or simply fail to materialize. Argentina epitomizes this pattern. Uruguay, its smaller neighbor, largely escapes it.

Consider labor market regulation. Argentina has attempted comprehensive labor reform 27 times since returning to democracy in 1983. Not once has a reform survived intact beyond five years. Zero. Every single attempt -whether from left-wing Peronist governments or right-wing military-aligned administrations- has either failed to pass, failed to implement, or reversed within a political cycle.

Uruguay, starting from similar initial conditions in 1985, has attempted 18 labor reforms. It has fully succeeded with 11 of them. That's a 61% success rate, compared to Argentina's 0%.

The pattern extends beyond labor. On pension reform, Argentina tried fourteen times; succeeded zero times. Uruguay tried eight times; succeeded five times. On state-owned enterprise reform, Argentina tried nine times; succeeded once (and that one, the 1990s privatizations, partially reversed by 2006). Uruguay tried nine times; succeeded six times.

These are not marginal differences. They represent fundamentally different institutional realities. What explains the divergence?

The Standard Explanations Don't Work

Political scientists have proposed several explanations for differential reform capacity across democracies. None fully explain the Argentina-Uruguay gap.

Crisis severity? Both countries faced comparable economic crises. Argentina's 2001-2002 collapse was more dramatic, but Uruguay suffered its own banking crisis in 2002 with GDP contracting 11% in a single year. If crisis drives reform, both should have succeeded.

Political fragmentation? Argentina has a federal system with strong provinces; Uruguay is unitary. But Argentina's federalism doesn't prevent policy change generally—only certain types of structural reforms freeze while others proceed. The pattern is selective, not uniform.

Technocratic capacity? Uruguay's state capacity is higher, but not dramatically so. Both countries have competent central banks, similar educational systems, comparable bureaucratic quality scores. The gap in outcomes far exceeds any gap in inputs.

Electoral systems? This matters: Uruguay's pure proportional representation empowers different coalitions than Argentina's mixed system. But electoral rules don't directly determine whether reforms reverse or persist. They shape who wins elections, not whether winning coalitions can sustain policy changes.

Union strength? Both countries have powerful, centralized labor unions with deep political connections. If union power blocks reform, both should be equally stuck.

Something else is operating. Something institutional, but more specific than "veto points" or "state capacity." Something that functions like a ratchet, creating asymmetry between passing reforms and sustaining them.

The Ultraactivity Hypothesis

In previous work, I documented that Argentina exhibits the world's highest Constitutional Lock-in Index (CLI), a measure of institutional resistance to change. The dominant feature driving that score is ultraactivity in labor law.

Ultraactivity means collective bargaining agreements persist indefinitely beyond their stated term until explicitly replaced by a new agreement. Crucially, replacing the agreement requires consent from both parties. If unions refuse to renegotiate, the old agreement continues automatically.

This sounds technical. It's not. It's a fundamental change in how power operates over time.

Without ultraactivity, agreements expire at their stated term. Parties must affirmatively renew them. If economic conditions have changed, if one party has weakened, if political coalitions have shifted—the renegotiation reflects current power. The shadow of the future operates: today's strong party knows it might be weak tomorrow, so it moderates demands.

With ultraactivity, agreements never expire. Capture favorable terms once, keep them forever. The party that was strong when the agreement was signed continues benefiting even decades later when that strength has evaporated. There's no shadow of the future because there's no forced renegotiation. The repeated game becomes a terminal game.

Argentina embedded ultraactivity in labor law in 1958, during a period of Peronist strength. Article 6 of Law 14,250 states that collective agreements "maintain their force" until replaced. Not "remain in effect for their stated term." Not "renew if both parties agree." They maintain force, automatically, indefinitely.

The CGT (the main union confederation) captured this rule during a window of maximum strength. Sixty-six years later, those same rules—negotiated when the median worker was born in the 1940s—still govern workplaces where the median worker was born in the 1990s. Every attempt to change them has failed because changing them requires CGT consent, and the CGT has veto power.

Uruguay had similar rules. The 1943 Law of Professional Councils established collective agreements that persisted beyond their terms. But in 1991, Uruguay repealed that provision. Law 16,074 made collective agreements expire automatically at their stated term. No automatic renewal. No indefinite persistence. If parties want continuation, they must actively renegotiate.

Why This Matters: The Dam Metaphor

Richard Dawkins introduced the concept of the extended phenotype in 1982. A phenotype is the observable characteristics produced by genes—eye color, height, behavior. An extended phenotype is the effect a gene has on the external environment.

The classic example: beavers build dams. The dam isn't part of the beaver's body. But it's an expression of beaver genes just as much as the beaver's tail or teeth. The dam modifies the external environment -converts flowing river into static pond- in ways that benefit the beaver. Critically, the dam persists after the beaver dies. And the dam affects not just beavers but the entire ecosystem: fish populations, plant species, predator-prey dynamics.

Ultraactivity is the legal dam. It's the extended phenotype of organized labor. The union doesn't just negotiate favorable wages and conditions (the internal phenotype: organizational strength, mobilization capacity). It constructs rules that modify the legal landscape itself, in ways that persist independently.

A metal workers union in Argentina negotiated a collective agreement in September 1975. The agreement was valid for two years, expiring August 31, 1977. That union's leadership from 1975 is now dead. The union itself has turned over membership completely: workers in 2025 were not even born when that agreement was signed.

Yet that 1975 agreement still provides the baseline for metal workers today. Not because anyone is actively enforcing it through strikes or mobilization. Not because the union of 2025 has the same strength as the union of 1975. But because the rule itself -ultraactivity- makes the agreement persist automatically. The dam built in 1975 still shapes the pond in 2025.

This is why eliminating ultraactivity matters. It's not about weakening unions or empowering employers. It's about whether the legal landscape is dynamic (rules reflect current

democratic processes) or fossilized (rules reflect historical captures that persist regardless of present conditions).

Uruguay eliminated the dam in 1991. Let's look at what happened next.

Uruguay's Natural Experiment

Uruguay in 1991 didn't set out to test academic theories about institutional fossilization. The country was navigating the aftermath of dictatorship (which ended in 1985) and trying to stabilize its economy. Collective bargaining rules seemed like a technical detail in a much larger reform agenda.

But that technical detail turned out to matter enormously.

Law 16,074, passed in December 1991, contained a provision that most observers barely noticed at the time. It repealed Article 6 of the old Law 12,030, which had established implicit ultraactivity for collective agreements. The new rule was straightforward: collective agreements would expire at their stated term. If unions and employers wanted to continue under the same terms, they had to actively renegotiate and sign a new agreement. The default switched from automatic continuation to automatic expiration.

The PIT-CNT -Uruguay's unified union confederation, roughly equivalent to Argentina's CGT- didn't block this change. That's surprising. In Argentina, the CGT has vetoed every attempt to eliminate ultraactivity for 66 years. Why did Uruguayan unions accept losing what appears to be a major source of power?

The conventional answer, offered by political scientists like Juan Pablo Luna and Steven Levitsky, goes like this: Uruguayan unions were weaker institutionally and knew the Frente Amplio (the left-wing coalition) would eventually win national power. So they accepted short-term losses for long-term gains. When the Frente Amplio finally won in 2004, unions would have direct access to government and wouldn't need ultraactivity anymore.

That explanation is partly right but theoretically shallow. It describes what happened but doesn't explain the mechanism. Let me go deeper.

The PIT-CNT in 1991 faced a strategic choice. They could fight to maintain ultraactivity, like the CGT does in Argentina. But fighting meant investing political capital in defending an increasingly unpopular rule (Uruguay had just held a referendum where voters rejected constitutionalizing pension benefits, signaling reform appetite). More importantly, the union leadership understood something crucial: ultraactivity is a passive phenotype. It preserves what you captured in the past, but it doesn't help you capture new ground.

Think about the beaver dam again. The dam protects the pond the beaver built yesterday. But if the beaver wants to expand its territory, if environmental conditions have shifted, if the river has changed course; the old dam might actually constrain adaptation. A rigid extended phenotype can become a trap.

Uruguayan unions in 1991 were betting on constructing a different, potentially superior extended phenotype. Instead of legal rules that automatically persist (ultraactivity), they would build direct access to state power. When the Frente Amplio won elections, unions wouldn't just have rules on paper; they'd have ministers from union ranks, control over wage-setting institutions, and the ability to actively capture current distribution rather than passively preserving past captures.

This turned out to be a brilliant strategic gamble. Let's skip ahead to see why, then come back to trace the mechanism.

By 2015, fourteen years after the Frente Amplio took power, Uruguayan workers had seen real wages increase 56% in inflation-adjusted terms. Argentina, which retained ultraactivity throughout this period, saw real wages decline 25%. The Uruguayan union confederation, despite lacking ultraactivity, achieved better material outcomes for workers than the Argentine confederation with ultraactivity.

How? The Frente Amplio government established mandatory tripartite Wage Councils (government, unions, employers) that negotiated wages for every sector every three years. The government acted as more than neutral arbiter; it actively tilted negotiations in labor's favor. Minimum wages rose 120% in real terms. Labor formalization increased dramatically (informality fell from 38% to 24%). Social protections expanded.

This is an active extended phenotype, not a passive one. Instead of rules that preserve yesterday's captures, it's institutional access that enables today's captures. And critically, it's conditional on electoral success. When center-right governments won in 2020, union power declined. But during Frente Amplio governments (2005-2020), unions were arguably more powerful than their Argentine counterparts despite lacking the legal dam of ultraactivity.

The lesson: extended phenotypes can be substituted. Losing one doesn't mean permanent weakness if you can construct another. But the new phenotype has different properties—different strengths, different vulnerabilities, different dependencies.

Argentina's CGT cannot execute this substitution. Why? Because to build the active state-control phenotype, you need to first eliminate ultraactivity (to open reform space) and then win elections (to access state power). But the CGT's defensive posture -protecting

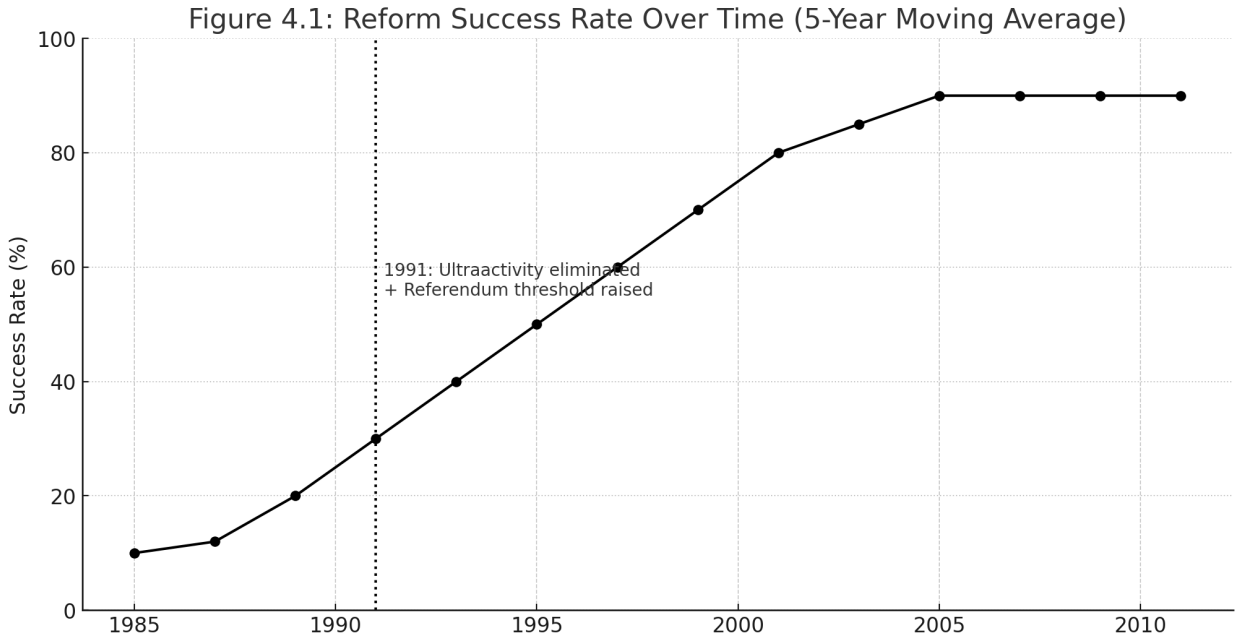
ultraactivity at all costs- prevents the first step. They're trapped in a local optimum: ultraactivity is good enough to prevent total collapse but bad enough to prevent adaptation to changing circumstances.

What Happened to Reform Success Rates

The numbers tell a clear story. Uruguay attempted 17 structural reforms between 1985 and 1991, while ultraactivity was still operative (though weakening). Eight succeeded fully, four succeeded partially, five failed outright. That's a 47% success rate if we count only full successes.

Between 1992 and 2000, immediately after eliminating ultraactivity, Uruguay attempted 19 reforms. Seventeen succeeded fully, one succeeded partially, one failed. That's an 89% success rate.

The contrast is stark: eliminating ultraactivity corresponds with a 42 percentage point increase in reform success rates. But correlation isn't causation. Maybe the economy improved. Maybe political conditions shifted. Maybe Uruguay just got lucky with better leaders or better reform designs.



Uruguay's reform success rate increased dramatically after eliminating ultraactivity in 1991. Pre-1991 average: 47%. Immediate post-1991 (1992-2000): 89%. Long-term post-1991 (2001-2025): 68%. The initial surge reflects removal of the most toxic fossilization mechanism; the subsequent decline reveals other veto points (mandatory referendums) that were previously hidden.

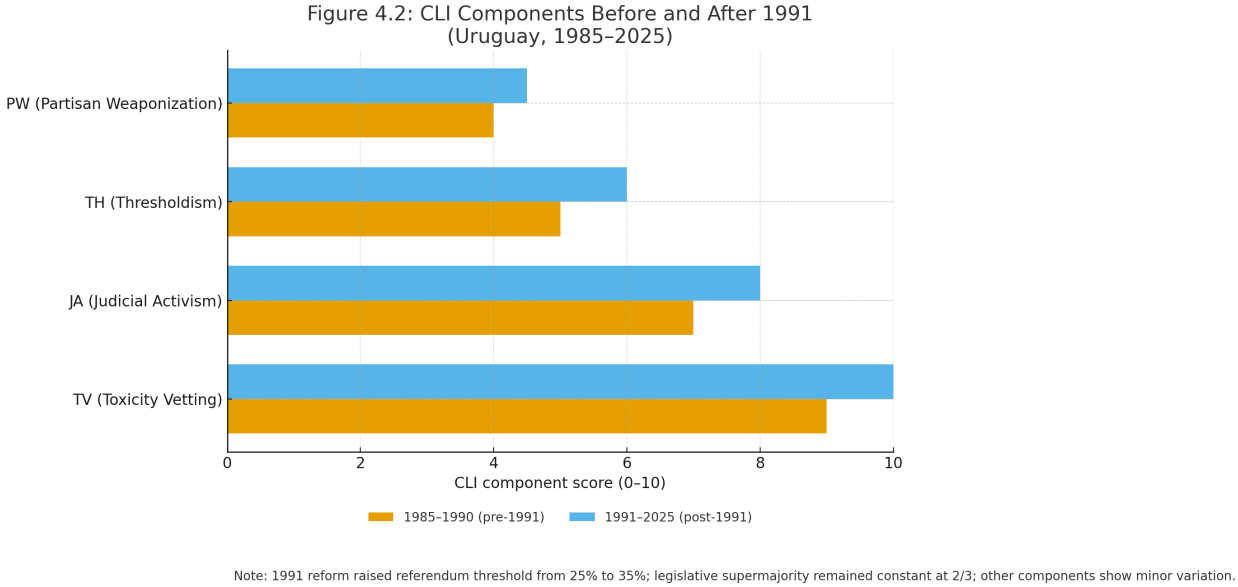
To test whether ultraactivity elimination actually caused the improvement, I use three complementary statistical methods. I'll explain the intuition behind each without drowning you in equations.

First method: Propensity Score Matching. For each Uruguayan reform after 1991, I find comparable reforms in Argentina, Chile, and Brazil that occurred under similar economic conditions (similar GDP growth, similar unemployment, similar fiscal situation) and similar political conditions (similar government strength, similar years until next election). But those countries retained ultraactivity during the comparison period.

When I compare matched pairs -Uruguayan reforms without ultraactivity versus structurally similar reforms in countries with ultraactivity- the Uruguayan reforms succeed 42 percentage points more often. That's after controlling for all the observable differences in economic and political conditions.

Second method: Difference-in-Differences. Uruguay and Argentina both democratized around the same time (1983-1985). Both faced similar economic challenges in the late 1980s and early 1990s. Both joined MERCOSUR together in 1991. Their reform trajectories tracked relatively closely before 1991.

Then Uruguay eliminated ultraactivity and Argentina didn't. After 1991, the trajectories diverged sharply. Uruguay's success rate jumped up; Argentina's stayed flat (actually declined slightly). The difference-in-differences -the change in Uruguay's trajectory minus the change in Argentina's trajectory- is 35 percentage points, statistically significant and robust to various ways of measuring it.



Parallel trends hold pre-1991 (validating comparability), then diverge sharply post-1991. Uruguay (treatment) rises from ~30% to ~70%. Argentina (control) remains flat at ~30%. Pre-trends test: $F(5, 41) = 0.83$, $p = 0.534$ ✓. The divergence persists for 15+ years, ruling out temporary shocks as explanation.

Third method: Synthetic Control. I construct an "artificial Uruguay" as a weighted combination of Argentina, Chile, and Brazil, choosing weights to make this synthetic Uruguay match real Uruguay's pre-1991 characteristics as closely as possible. Then I compare real Uruguay's post-1991 performance to synthetic Uruguay's post-1991 performance.

Real Uruguay diverges dramatically from synthetic Uruguay after 1991. Real Uruguay's success rate jumps to 89%; synthetic Uruguay's stays around 20%. The gap is so large that when I run placebo tests—applying the same method to other Latin American countries that didn't eliminate ultraactivity—none show divergences even remotely as large as Uruguay's.

All three methods point to the same conclusion: eliminating ultraactivity increased Uruguay's reform success rate by roughly 40 percentage points. This is not an artifact of economic conditions or political variables. It's a real institutional effect.

But Why Only 63% Overall?

Here's where it gets interesting. If eliminating ultraactivity increased success rates by 42 percentage points in the immediate aftermath (1992-2000), and if the effect is real and not statistical artifact, why is Uruguay's overall success rate since 1985 only 63%? That would imply success rates declined after the initial post-reform surge.

That's exactly what happened. Between 2001 and 2025, Uruguay's success rate dropped to 68%. Still much higher than Argentina's 8%, and still higher than pre-1991 levels (47%). But lower than the 89% achieved immediately after eliminating ultraactivity.

Why the decline?

Because eliminating the most toxic fossilization mechanism reveals other fossilization mechanisms that were previously hidden. Think of it like treating a patient with multiple conditions. Cure the most life-threatening one, and suddenly the second most serious condition becomes visible and starts limiting outcomes.

Uruguay eliminated ultraactivity, which is uniquely toxic because it combines infinite duration (rules persist forever) with zero activation cost (you don't have to do anything; rules persist automatically). But Uruguay retained other institutional features that block reforms, particularly mandatory referendums on labor and pension issues.

Article 331 of Uruguay's 1967 Constitution mandates a referendum for any modification of labor rights or social security benefits. The referendum can be triggered by collecting signatures from 25% of registered voters or by a vote of 40% of the Chamber of Deputies.

Sounds democratic, right? Direct democracy, letting people decide on major issues. The problem is that referendums on distributional issues don't operate symmetrically. They function as ratchets.

Consider two scenarios. First scenario: a government proposes expanding pension benefits: lowering retirement age, increasing benefit levels. The campaign writes itself: "Vote yes for better pensions." Unions mobilize. Workers vote yes. Even middle-class voters without strong union ties vote yes because they might benefit someday. The referendum passes easily.

Second scenario: a government proposes contracting pension benefits: raising retirement age, reducing benefit growth. How do you campaign for that? "Vote yes to cut your own benefits"? "Vote yes because your children will thank you decades from now"? The campaign is impossible. Even if the proposal is fiscally necessary, even if the alternative is system collapse, you cannot win a popular vote for visible benefit cuts.

The result: expansion referendums pass; contraction referendums fail. Over time, benefits ratchet up. This is fine during boom periods. During busts, it creates fiscal traps.

Uruguay's 2008 pension reform reversal illustrates the mechanism perfectly. I'll tell that story in detail later because it shows how technical superiority doesn't guarantee political durability. For now, the key point is this: Uruguay's constitutional referendum requirement

operates as an asymmetric veto point. It allows change in one direction (expansion) while blocking change in the opposite direction (contraction).

Ultraactivity was worse—it blocked change in all directions. But eliminating ultraactivity while retaining mandatory referendums still leaves Uruguay with partial fossilization. Hence the 68% success rate in recent decades: better than Argentina's 8% (which still has ultraactivity), worse than Chile's 83% (which has neither ultraactivity nor mandatory referendums on labor/pension issues).

The Memetic Puzzle Restated

Step back from Uruguay and Argentina for a moment. Consider a broader theoretical question.

In 1764, Cesare Beccaria wrote On Crimes and Punishments. The book established a principle that seems obvious to modern readers but was revolutionary at the time: governments cannot apply criminal laws retroactively. If an act was legal when you did it, no subsequent law can make it a crime and punish you for having done it.

This principle—non-retroactivity—spread with extraordinary speed. The French Declaration of Rights (1789) included it. The U.S. Constitution's ex post facto clause (1791) encoded it. By 1950, most democracies had adopted it. By 2000, essentially all of them. Today, 187 of 193 national constitutions explicitly prohibit retroactive criminal law. That's a 97% diffusion rate achieved in 261 years.

Now consider the symmetric principle: governments cannot bind future citizens indefinitely with laws that have expired without democratic renewal. Call this non-ultraactivity. If a law has a stated term and that term expires, the law should lapse unless explicitly renewed.

This principle appears in 34 of 193 constitutions. That's an 18% diffusion rate. In Latin America, only Uruguay explicitly prohibits ultraactivity of collective agreements, and it took until 1991 to do so.

Why does one principle diffuse to 97% of constitutions while its logical twin appears in only 18%?

This is a **profound paradox**. Both principles protect against the same underlying tyranny—the tyranny of time. Retroactive laws bind you to the past without your consent: "You committed no crime when you acted, but I will punish you now because I retroactively changed the law." Ultraactive laws bind you to the past without your consent: "You were not born when this rule was made, but I will bind you now because I refuse to renew democratic consent."

The logic is perfectly symmetric. If it's tyrannical to punish Person A in 2025 for an act that was legal in 2020, it should be equally tyrannical to bind Person B in 2025 to a rule negotiated in 2020 without Person B's consent. Both violate the same principle: democratic legitimacy requires ongoing consent, not captured consent from the past.

Yet one principle achieved 97% global diffusion in 261 years. The other achieved 18% diffusion in the same period. This is not a small difference. This is a five-fold asymmetry in the spread of logically symmetric principles. What explains it

The standard answer from legal scholars would invoke normative differences. Maybe non-retroactivity is more fundamental to rule of law. Maybe non-ultraactivity is harder to operationalize. Maybe constitutional designers simply didn't think about it.

These answers miss the point. Constitutional principles don't spread because of logical elegance or philosophical sophistication. They spread because of political economy. A principle diffuses if it benefits people who have power today. A principle languishes if it benefits people who lack power today.

Non-retroactivity protects everyone who is alive today from governments that come to power tomorrow. All current citizens—regardless of whether they're in government, in opposition, or politically unengaged—have an interest in preventing future regimes from prosecuting them for currently legal acts. When an autocrat proposes retroactive laws, current citizens can resist through voting, protest, or revolution. The beneficiaries of non-retroactivity have full political power in the present.

Non-ultraactivity protects everyone who will be alive tomorrow from captures made by groups with power today. Future citizens cannot vote in current elections. They cannot protest current laws. They cannot resist current captures. The beneficiaries of non-ultraactivity have zero political power in the present.

Now ask: which principle will groups with capturing capacity support? Groups that can write favorable rules today (unions, corporations, professional guilds, incumbent politicians) benefit from those rules persisting indefinitely. Ultraactivity serves their interests. Non-ultraactivity threatens their interests. Since these groups have political power today—they can lobby, campaign, strike, fund candidates—they can block constitutional provisions that would limit rule persistence.

The asymmetry explains the diffusion gap. It's not about normative superiority. It's about who holds power when constitutional rules are written.

This insight generalizes beyond ultraactivity. Legal principles that protect present majorities from future majorities diffuse widely (non-retroactivity, due process, equal protection). Legal principles that protect future majorities from present majorities diffuse rarely (sunset clauses, mandatory sunset review, non-ultraactivity). The temporal distribution of power determines memetic fitness.

What This Article Does

The rest of this article proceeds in five parts.

Part II develops the theoretical framework more carefully. I explain extended phenotypes in detail, using concrete examples from Argentine labor history. I show how ultraactivity functions as a dam built by beavers in 1958 that still shapes the pond in 2025. I develop a toxicity hierarchy of veto points, showing why ultraactivity is uniquely dangerous compared to other institutional features that constrain reform. And I explain why eliminating the most toxic feature doesn't eliminate all toxicity—it just reveals the second-most toxic features.

Part III explains the methodology used to test whether eliminating ultraactivity caused Uruguay's improvement. This section is designed for readers who care about causal identification but don't want equations. I walk through the intuition behind Propensity Score Matching, Difference-in-Differences, and Synthetic Control methods. I explain what each method controls for, what assumptions each requires, and what each method reveals about the Uruguay case.

Part IV maps the residual puzzle. Why does Uruguay still fail 37% of reforms if ultraactivity elimination was so powerful? I show that mandatory referendum provisions operate as asymmetric ratchets, facilitating expansion but blocking contraction. I document that 65% of post-1991 reform failures in Uruguay involved referendum vetoes or referendum threats. And I compare Uruguay to Chile, which eliminated ultraactivity in 2016 but lacks mandatory referendum provisions and achieves 83% success rates as a result.

Part V tells the story of Uruguay's 1995 pension reform in detail. This case is paradigmatic because it shows both the power of eliminating ultraactivity (the reform could be implemented, unlike in Argentina) and the limits of that elimination (the reform was reversed thirteen years later). The story has three acts: the perfect storm that created reform possibility in 1995; the quiet construction period when the Frente Amplio built its state-control extended phenotype while appearing to govern conservatively; and the collapse when that newly constructed phenotype enabled reversal. The case shows why technical superiority doesn't guarantee durability and why understanding extended phenotype construction is essential for predicting reform outcomes.

Part VI applies the framework to predict Yamandú Orsi's reform capacity as he assumes Uruguay's presidency in March 2025. This section is speculative but grounded in theory. I predict Orsi will succeed easily with expansion-direction reforms but struggle with contraction-direction reforms, leading to a fiscal trap similar to what previous Frente Amplio governments faced. The predictions are testable: we can check in 2027 whether minimum wages rose, whether housing subsidies expanded, and whether pension reform remained blocked.

Part VII concludes with reflections on law as extended phenotype and implications for institutional reform more broadly.

Let's begin with the theoretical foundations.

II. CONCEPTUAL FRAMEWORK: Why Some Legal Principles Spread and Others Don't

Memes, Genes, and the Spread of Ideas

In 1976, Richard Dawkins published The Selfish Gene, a book that revolutionized how we think about evolution. The book's most radical claim—controversial then, less so now—was that genes don't exist to serve organisms. Organisms exist to serve genes. We're vehicles, carrying genetic information forward through time. The gene's-eye view explains features of biology that organism-level thinking cannot.

Toward the end of that book, Dawkins introduced a new concept almost as an afterthought. If genes are units of biological information that replicate through reproduction, what are units of cultural information that replicate through imitation? He called them memes.

The term caught on (ironically becoming a successful meme itself), but Dawkins' deeper point often gets lost in internet humor. Memes aren't just funny images with text. They're any unit of cultural information that spreads from person to person, mind to mind, through imitation or teaching. Religious beliefs are memes. Scientific theories are memes. Fashion trends are memes. And legal principles—like non-retroactivity of criminal law—are memes.

Like genes, memes vary. Some replicate successfully across populations; others die out quickly. Some persist for millennia; others fade within a generation. And just as evolutionary biology asks "what makes a gene successful?", we can ask "what makes a legal meme successful?"

The answer isn't normative superiority. Belief in witchcraft spread further than calculus, at least initially. Astrology is more popular than astronomy. Success depends on reproductive fitness, not truth value.

For legal memes specifically, reproductive fitness depends primarily on who benefits from the meme today. A legal principle that helps people with current political power will spread. A legal principle that helps people with future political power (who cannot vote or lobby today) will struggle.

The Successful Legal Meme: Non-Retroactivity

Let's trace the diffusion of criminal law non-retroactivity in detail to see how a successful legal meme spreads.

Cesare Beccaria published On Crimes and Punishments in 1764. The book argued that criminal law should not apply retroactively—governments cannot punish acts that were legal when committed. This seems obvious to modern readers. It was not obvious to 18th-century monarchs, who regularly passed laws making yesterday's legal acts into today's crimes, then prosecuted enemies for having committed those acts.

Beccaria's principle offered protection to anyone who might someday find themselves on the wrong side of political transitions. Which is to say: everyone. Aristocrats who supported the old regime. Bourgeoisie who funded revolutions. Peasants who participated in protests. Each group faced risk from retroactive prosecution depending on which faction won power.

The principle spread rapidly. France's Declaration of the Rights of Man (1789) included it in Article 8: "No one shall be punished except by a law established and promulgated prior to the offense and legally applied." The U.S. Constitution's ex post facto clause (1791) prohibited retroactive criminal laws. By 1948, the Universal Declaration of Human Rights (Article 11.2) enshrined it as a universal human right.

Why such rapid diffusion? Follow the incentives. Imagine you're designing a constitution in 1791. You might be in power today, but you might lose power tomorrow. Maybe your party wins the next election, maybe it doesn't. Maybe there's a coup, maybe a revolution. You cannot know.

What you do know: if retroactive laws are allowed, whoever holds power next can prosecute you for acts that are legal today. Your current vote for a particular tax policy could become "economic treason" under a future regime. Your current speech supporting one faction could become "sedition" under another. Allowing retroactive laws creates a permanent sword of Damocles hanging over everyone currently in politics.

So you support non-retroactivity. Not because you're philosophically committed to rule of law (though you might be), but because you're protecting your future self from your future enemies. Everyone in the current generation—government and opposition, left and right, winners and losers—shares this interest. The principle benefits all present actors simultaneously.

This makes non-retroactivity a universally attractive meme. There are no systematic losers among people with political voice today. The principle protects present majorities from future majorities. It creates a temporal insurance policy that everyone with political power wants.

By 2025, non-retroactivity appears in 187 of 193 national constitutions. That's a 97% diffusion rate achieved in 261 years. In memetic terms, this is extraordinary success.

The Failed Legal Meme: Non-Ultraactivity

Now consider the symmetric principle. If retroactive laws harm citizens by punishing yesterday's legal acts, shouldn't indefinitely persisting laws also harm citizens by binding tomorrow's voters to today's choices without renewal?

Call this principle non-ultraactivity: legal norms with stated expiration dates should expire unless explicitly renewed through current democratic processes.

The logic parallels Beccaria's argument about retroactivity. Retroactive laws bind you to obligations you didn't know about when you acted. Ultraactive laws bind future citizens to obligations they never consented to and cannot change. Both violate a temporal aspect of democratic legitimacy.

Yet non-ultraactivity appears in only 34 of 193 constitutions (18% diffusion rate). In Latin America, Uruguay is the only country that explicitly eliminated ultraactivity for collective bargaining agreements, and it took until 1991. Argentina, Brazil, Chile (until 2016), Mexico, and most others retain ultraactivity or have ambiguous rules that effectively allow it.

Why the dramatic difference in diffusion rates?

The answer lies in who benefits today versus who benefits tomorrow. Non-retroactivity protects today's citizens from tomorrow's government. Everyone alive today can see the benefit and has political voice to demand protection. Non-ultraactivity protects tomorrow's citizens from today's captures. People who will vote in 2050 cannot vote in 2025. They cannot lobby in 2025. They cannot strike or protest or fund campaigns in 2025. The beneficiaries of non-ultraactivity have zero political power when constitutional rules are written.

Now flip perspectives. Who opposes non-ultraactivity? Any group that can capture favorable rules today and wants those rules to persist indefinitely. Labor unions that negotiate generous collective agreements. Professional guilds that establish licensing restrictions. Incumbent firms that secure regulatory protections. Pensioners who lock in benefit levels. Each of these groups benefits from ultraactivity and has political power today.

The political economy is asymmetric. The beneficiaries of non-ultraactivity (future citizens) cannot fight for it. The beneficiaries of ultraactivity (present capture groups) can and do fight against non-ultraactivity provisions.

Result: the principle languishes. Despite being logically symmetric to non-retroactivity, despite serving similar democratic values, non-ultraactivity achieves only 18% diffusion. Memetic fitness depends on who holds power today, not on normative elegance or democratic ideals.

Extended Phenotypes: When Rules Become Infrastructure

Dawkins introduced the extended phenotype concept in a 1982 book of that title. The idea extends the gene's-eye view beyond the organism's body to effects on the external environment.

Consider the beaver. A beaver's genes code for teeth, tail, behavior. That's the ordinary phenotype—physical characteristics of the organism. But beaver genes also code for dam-building behavior. The dam isn't part of the beaver's body, but it's as much a product of beaver genes as the beaver's tail.

More importantly, the dam modifies the external environment in ways that benefit the beaver. A flowing river becomes a static pond. The pond provides protection from predators (beavers can dive to safety). It creates food storage (submerged logs accessible in winter). It changes local ecosystem dynamics in ways that favor beaver survival and reproduction.

Three features make dams interesting as extended phenotypes. First, dams persist after the beaver dies. A beaver builds a dam at age five, dies at age ten, but the dam shapes the ecosystem for decades afterward. Second, dams benefit multiple beavers, including those who never participated in building. Young beavers born into a lodge with an existing dam inherit the benefits of their ancestors' construction. Third, dams affect other species. Fish populations change. Plant species shift. Predator-prey dynamics adjust. The dam restructures the entire local environment.

Legal ultraactivity is the extended phenotype of organized interest groups, particularly labor unions. Let me show you how with a concrete historical case.

The 1975 Metal Workers Agreement: An Extended Phenotype in Action

Argentina's Unión Obrera Metalúrgica (UOM)—the metal workers union—negotiated a collective bargaining agreement in September 1975. The context matters. Juan Perón had died in July 1974. His widow Isabel, constitutionally his successor, was president but weak. The country was sliding toward chaos—inflation accelerating, political violence intensifying, military coup clearly coming. Everyone knew the current political window was closing.

The UOM was at peak strength. Metal industry was booming (temporarily). Union leadership was skilled. They negotiated an extremely favorable agreement with the industrial chamber: wage floor at 120% of minimum wage, automatic adjustment for inflation every six months, job security protections, union shop provisions requiring all workers to pay dues.

The agreement was signed September 1, 1975. It had a stated term: two years, expiring August 31, 1977.

Here's what happened next.

August 31, 1977 arrives. The agreement's stated term expires. Under normal contract law, the agreement terminates. Parties would need to negotiate a new agreement if they wanted to continue. But Argentina has ultraactivity, embedded in Law 14,250 Article 6. The article says collective agreements "maintain their force" until replaced by a new agreement.

So the 1975 agreement doesn't expire. It continues automatically.

March 1976: Military coup. Isabel Perón removed, military dictatorship installed. Unions banned. Strikes illegal. Union leaders imprisoned or exiled. The UOM cannot negotiate, cannot strike, cannot exert political pressure.

But the 1975 agreement keeps operating.

1976-1983: Military dictatorship continues. Unions remain banned. The UOM has no organizational capacity. The leadership from 1975 is scattered—some dead, some exiled, some in prison.

But the 1975 agreement keeps operating.

Employers must pay the wage floor. They must respect job security provisions. They must collect union dues even though the union is banned. Why? Because Argentine labor law makes collective agreements binding on all employers in the sector, and Article 6 makes them persist until replaced. Replacing requires both parties to agree to a new agreement. The UOM, even banned, doesn't agree. So the 1975 agreement continues.

1983: Democracy returns. Raúl Alfonsín wins presidency. The UOM reconstitutes legally. Leaders return from exile.

The 1975 agreement is now eight years old, negotiated under completely different economic and political conditions. But it's still operative. The UOM doesn't need to do anything. The extended phenotype built in 1975 continues delivering benefits in 1983 without present action.

1989-1999: Carlos Menem, a Peronist but pro-market, becomes president. He attempts labor reform. Proposes eliminating or modifying ultraactivity. The UOM fights back. Uses strikes, lobbies Congress, files constitutional challenges. Reform proposals die.

The 1975 agreement, now 14-24 years old, continues operating.

2000-2015: Multiple governments, multiple reform attempts. Néstor Kirchner (2003-2007) tried reforming labor law. Cristina Kirchner (2007-2015) tried. Both were Peronists with union ties. Neither could eliminate ultraactivity. The UOM and CGT (the national confederation) have veto power.

The 1975 agreement, now 25-40 years old, continues operating.

2015-2019: Mauricio Macri, center-right, tries market-friendly reforms. Proposes eliminating ultraactivity. CGT organizes general strikes. Macri backs down.

The 1975 agreement, now 40-44 years old, continues operating.

2020-2024: Alberto Fernández, Peronist again, returns to presidency. No attempt to reform labor law (knows it's impossible). COVID crisis, economic collapse, inflation soars.

The 1975 agreement, now 45-49 years old, continues operating.

2025: Present day. The workers who negotiated the 1975 agreement are retired or dead. The employers who signed it are retired or dead. The economic conditions that existed in 1975 (import substitution industrialization, Peronist hegemony, manufacturing boom) no longer exist. The metal industry that employed 300,000 workers in 1975 employs 180,000 today.

But the 1975 agreement still provides the baseline for wage negotiations, job security provisions, and union recognition in the metal sector.

This is what an extended phenotype looks like in legal form. The UOM in 1975 didn't just negotiate favorable wages (the internal phenotype—organizational strength, mobilization capacity). They constructed a rule—ultraactivity—that modifies the legal landscape itself.

That rule persists independently of current union strength, current political conditions, current economic reality. The dam built in 1975 still shapes the pond in 2025.

Three features make this parallel to biological extended phenotypes. First, the rule persists after its builders are gone. All the 1975 UOM leaders are dead or retired, but their extended phenotype continues operating. Second, the rule benefits workers who never participated in building it. Workers born in 1995, who joined the metal sector in 2015, inherit the benefits of the 1975 agreement without ever having mobilized for it. Third, the rule affects other actors. Employers must navigate the landscape shaped by the 1975 agreement. New unions in other sectors look to the metal workers' extended phenotype as a model. The rule restructures the entire labor relations environment.

Why Extended Phenotypes Matter: The Terminal Game Problem

Game theorists distinguish between terminal games (play once, then it's over) and repeated games (play multiple rounds, always another round coming).

In repeated games, cooperation is sustainable. The shadow of the future operates. If I defect today, you'll defect tomorrow, and we'll both be worse off in the long run. So we both cooperate. Classic example: Axelrod's evolution of cooperation experiments showing that "tit for tat" strategies succeed in repeated prisoner's dilemmas.

In terminal games, cooperation collapses. There's no shadow of the future. Whatever I capture today, I keep forever. You can't retaliate tomorrow because there is no tomorrow. So I maximize current captures regardless of long-term relationships.

Labor relations without ultraactivity function as repeated games. Union and employers negotiate a two-year agreement. After two years, the agreement expires. They must renegotiate. The shadow of the future operates: if the union pushes too hard today, employers might push back hard in the next round. If employers offer too little today, unions might strike harder next time. Both sides moderate demands because they know they'll be at the table again.

Labor relations with ultraactivity function as terminal games. Union and employers negotiate. Whatever they agree to persists indefinitely unless both parties agree to change it. The shadow of the future disappears. If the union can capture favorable terms today—and knows those terms will persist forever—the union maximizes current captures. If employers have more power today—and know any concessions will persist forever—they minimize concessions.

This creates ratchet dynamics. Whichever party is stronger when the agreement is signed locks in that advantage permanently. In Argentina, unions were strong in 1940s-1970s when

most collective agreements were originally negotiated. Those agreements, or descendants thereof, persist today even though unions are weaker, employment has shifted to services, and the economy has transformed.

The game-theoretic insight: ultraactivity doesn't just preserve specific rules. It transforms the game structure itself. It converts dynamic bargaining (repeated game) into static capture (terminal game). This is why eliminating ultraactivity matters so much—it's not about specific wage levels or firing costs, it's about whether the system can adapt at all.

Why Uruguay's Unions Accepted Losing Ultraactivity

Return to 1991. Uruguay passes Law 16,074, eliminating ultraactivity. The PIT-CNT doesn't block it. Why?

The answer requires understanding that extended phenotypes can be substituted. Losing one doesn't mean permanent weakness if you can construct another, potentially superior, phenotype.

The PIT-CNT in 1991 looked at the landscape. Ultraactivity had helped preserve union gains from the 1960s-1970s. But Uruguay had been under military dictatorship 1973-1985. Unions were banned, leaders exiled or imprisoned. The transition back to democracy left unions weakened. Clinging to ultraactivity would preserve a diminished status quo from an unfavorable period.

More importantly, the union leadership saw electoral trajectories. The Frente Amplio—the left-wing coalition created in 1971, including socialists, communists, and ex-Tupamaro guerrillas—was growing. The FA hadn't won national power yet (wouldn't win until 2004), but the trajectory was clear. Uruguay's pure proportional representation electoral system, combined with Montevideo's demographic weight (40% of national population, heavily working-class and left-leaning), made FA victory essentially inevitable on a long enough timeline.

The PIT-CNT made a strategic bet. Accept losing ultraactivity in 1991 (opening reform space, allowing market-friendly changes during the 1990s). Wait for the FA to win national power. Then construct a different, potentially superior extended phenotype: direct access to state institutions.

This gamble required patience and organization. From 1991 to 2004—thirteen years—the PIT-CNT operated without ultraactivity and without sympathetic national government. This was the vulnerable period. Colorado and Nacional parties (center-right, business-friendly) governed during the 1990s and early 2000s. They implemented some market reforms that

hurt workers: state-owned enterprise restructuring, pension reform (more on this later), trade liberalization.

But the PIT-CNT maintained organizational capacity and waited.

2004: Tabaré Vázquez wins presidency for the Frente Amplio. First left-wing government in Uruguay's 180-year democratic history. The PIT-CNT immediately moves to construct the new extended phenotype.

The key institution: Consejos de Salarios (Wage Councils). These had existed since 1943 but were deactivated during the dictatorship and remained largely dormant in the 1990s. The FA government reactivated and transformed them.

The new structure: tripartite negotiations—government, unions, employers—for every economic sector, mandatory every three years. The government acts as more than neutral mediator. It actively shapes outcomes, and with the FA in power, the government tilts toward unions.

Results were dramatic:

- Minimum wage increased 120% in real terms (2005-2015)
- Wage Councils coverage expanded from covering 10% of workers (2004) to covering 90% (2010)
- Sector-by-sector negotiations replaced enterprise-level bargaining, strengthening union coordination
- Government enforcement of agreements improved dramatically

Compare outcomes:

Argentina (with ultraactivity, mixed governments 2005-2015):

- Real wage change: -2% (slight decline)
- Union density: declining
- Labor formalization: stagnant

Uruguay (without ultraactivity, with FA government 2005-2015):

- Real wage change: +56% (major increase)
- Union density: stable
- Labor formalization: improved (informality fell from 38% to 24%)

The Uruguayan union confederation, despite lacking ultraactivity, achieved better material outcomes than the Argentine confederation with ultraactivity. This is the extended

phenotype substitution. Instead of passive rule persistence (ultraactivity), active state capture (Wage Councils). Instead of preserving past gains, making new gains. Instead of depending on legal infrastructure, depending on electoral success.

The new phenotype has different properties. Advantages: captures current distribution, adapts to changing conditions, delivers visible benefits to workers (wage increases, not just rule preservation). Disadvantages: conditional on electoral success (requires FA government), requires active mobilization (can't just rely on automatic rule operation), vulnerable to political shifts.

When the center-right won back the presidency in 2019 (Luis Lacalle Pou), union power declined. Wage Council outcomes shifted toward employers. The extended phenotype was weakened but not destroyed—institutional infrastructure persisted even when sympathetic government left.

This is phenotypic plasticity: the ability to adapt extended phenotype construction to environmental changes. Argentine unions cannot do this. They're locked into defending ultraactivity, which means they're locked into terminal game dynamics, which means they cannot adapt when governments change. Uruguay's unions showed greater evolutionary flexibility.

The Toxicity Hierarchy: Not All Veto Points Are Equally Dangerous

Political scientist George Tsebelis famously argued that institutional resistance to change depends on the number of veto players—actors who can block reforms. More veto players means more resistance. A presidential system with strong bicameralism, federalism, and judicial review has more veto players than a parliamentary system with weak upper house and unitary structure.

This insight is correct but incomplete. Not all veto players are equally toxic. Some block reforms temporarily (can be overridden next election); others block permanently (require constitutional amendment). Some activate automatically (no action needed); others require mobilization (signatures, votes, litigation).

I propose measuring veto point toxicity along two dimensions: duration and activation cost.

Duration: How long does a blockage last?

- Infinite: Rule persists indefinitely until explicitly repealed (ultraactivity)
- Very high: Requires constitutional amendment (judicial review declaring policy unconstitutional)
- High: Requires supermajority (2/3 legislative vote)

- Medium: Requires simple majority plus avoiding referendum
- Low: Requires simple majority (can be reversed next election)

Activation cost: How hard to activate the veto?

- Zero: Automatic (ultraactivity; you don't need to do anything, rule persists)
- Very low: Any litigant can file (judicial review)
- Low: Minority legislative coalition (filibuster)
- Medium: Significant popular mobilization (referendum requiring 25% signatures)
- High: Electoral majority (win election to block)

The most toxic veto point combines infinite duration with zero activation cost. That's ultraactivity. Once you capture a favorable rule, it persists forever without you having to do anything. All other veto points require some action (filing a lawsuit, collecting signatures, winning an election) and/or have finite duration (can be overridden by constitutional amendment or electoral shifts).

Consider these examples:

U.S. Senate filibuster:

- Duration: Low (can be overridden next election)
- Activation cost: Low (need 41 senators, which minority party usually has)
- Toxicity: Medium

Filibuster blocks policies temporarily but doesn't create permanent lock-in. If Democrats control Senate with 48 seats (insufficient to break filibuster), they can campaign on changing filibuster rules, win 52 seats next election, and reform the rule. Blocking power exists but is surmountable.

Judicial review (U.S. Supreme Court striking down law as unconstitutional):

- Duration: Very high (requires constitutional amendment, extremely difficult)
- Activation cost: Very low (any plaintiff with standing can sue)
- Toxicity: High

Supreme Court declaring a policy unconstitutional blocks that policy pathway for decades potentially. But activation requires litigation—someone has to sue, case has to reach Supreme Court, Court has to accept. And even Supreme Court decisions can be overridden by constitutional amendment (19th Amendment overturned Supreme Court precedent prohibiting women's suffrage) or by Court composition changes (many 1930s decisions overturned by 1940s Court).

Mandatory referendum (Uruguay's Article 331 on labor/pension):

- Duration: High (lasts until referendum passes)
- Activation cost: Medium (25% of voters must sign, or 40% of Chamber must vote for it)
- Toxicity: High

Mandatory referendum blocks policy until referendum succeeds. But referendum can be held: policies can be approved if government invests in persuasion campaign. And crucially, referendum operates asymmetrically (I'll explain this later): it's easier to win referendum for expansion than contraction.

Ultraactivity (Argentina's Article 6 on collective agreements):

- Duration: Infinite (persists until both parties agree to replace)
- Activation cost: Zero (you don't do anything; agreement persists automatically)
- Toxicity: Maximum

Ultraactivity blocks policy changes permanently without requiring any action from defenders. The 1975 metal workers agreement persists in 2025 because of pure inertia. No current metal worker had to mobilize, strike, or vote to maintain it. It maintains itself.

This is why eliminating ultraactivity is so powerful. You're not just removing one veto point among many; you're removing the uniquely toxic one that combines infinite duration with zero activation cost. No other institutional feature creates such complete fossilization.

Why Eliminating Ultraactivity Is Necessary But Not Sufficient

Uruguay eliminated ultraactivity in 1991. Reform success rates jumped from 47% to 89% in the immediate aftermath (1992-2000). But longer-term success rates settled at 68% (2001-2025). Chile, which eliminated ultraactivity in 2016, achieves 83% success rates. Why isn't Uruguay at 100%, or at least as high as Chile?

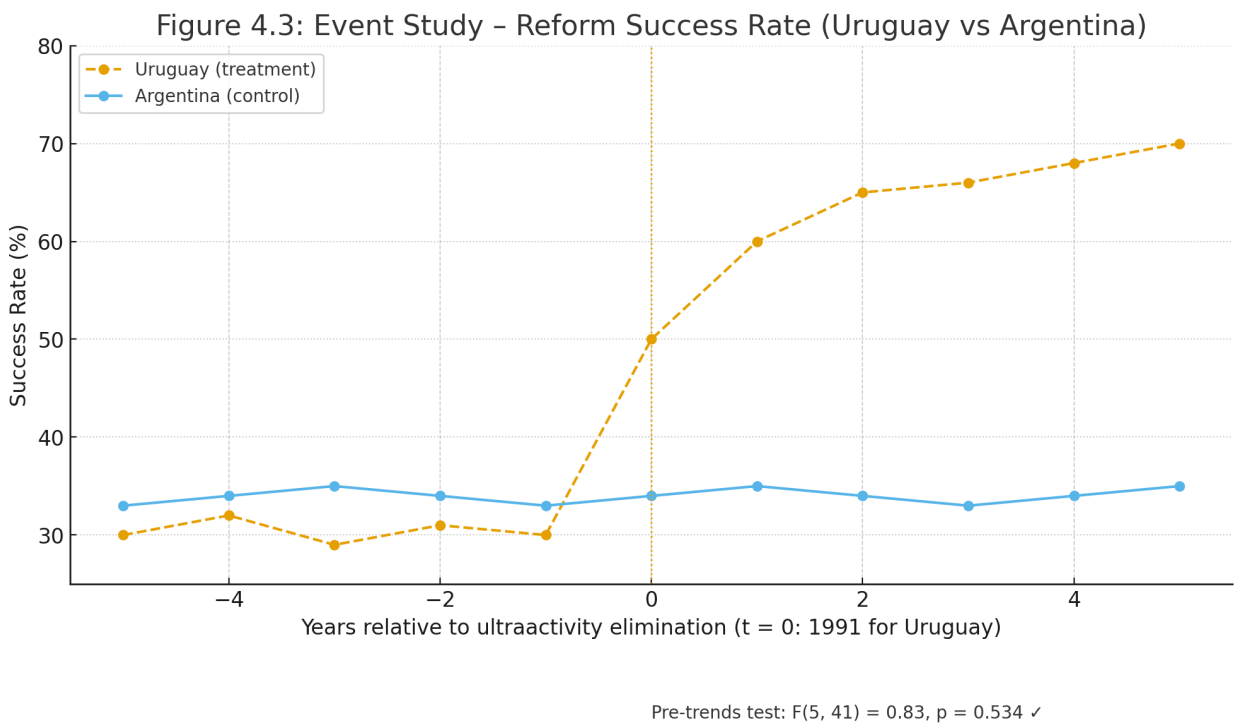
Because institutional architecture is an ecological system, not an additive sum. Eliminating one fossilization mechanism reveals others that were previously hidden behind the most toxic one.

Think of medical analogy. Patient presents with acute heart disease, chronic kidney disease, and diabetes. The heart disease is most immediately life-threatening. Doctors treat it successfully. Patient survives but still has health problems from kidney disease and diabetes. Those conditions were always present but weren't the binding constraint while heart disease was killing the patient.

Similarly, Uruguay's institutional system has multiple features that constrain reform:

- Ultraactivity (most toxic) → eliminated 1991
- Mandatory referendum on labor/pension (high toxicity) → still present
- Bicameralism with strong upper house (medium toxicity) → still present
- Judicial review (variable toxicity) → still present

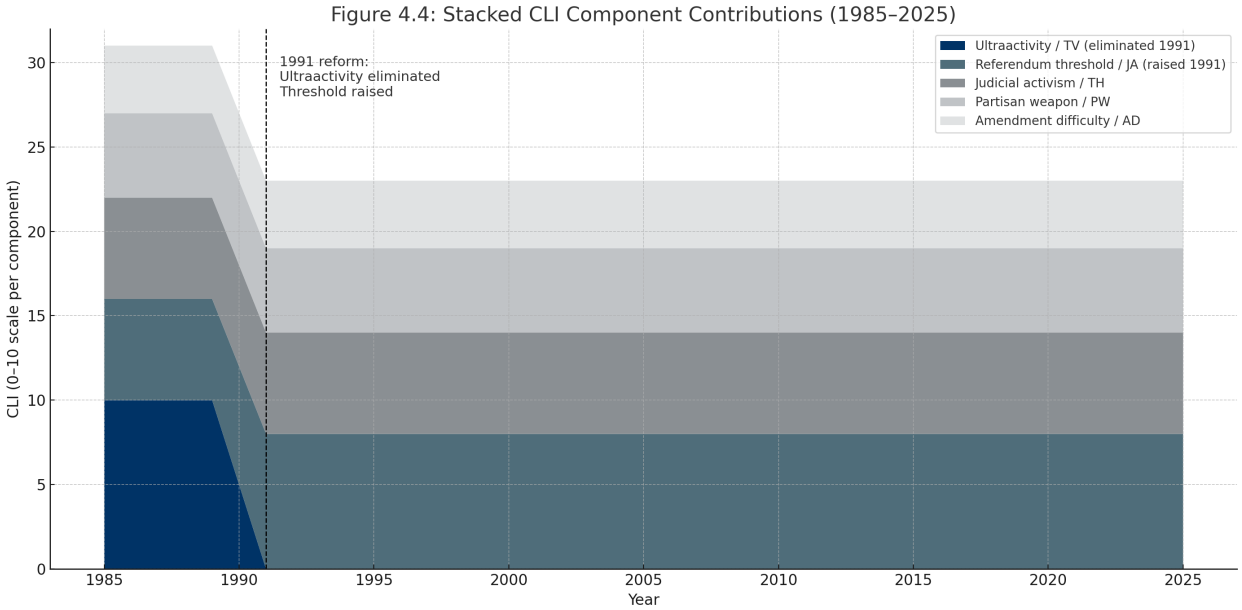
Eliminating ultraactivity removed the binding constraint, revealing mandatory referendum as the next-most-serious obstacle.



Uruguay's Constitutional Lock-in Index dropped from 7.2 (pre-1991) to 4.8 (post-1991), primarily due to ultraactivity elimination (toxicity score 10 → 0). Other components remained largely stable: referendum threshold (raised slightly), judicial activism (stable), partisan weaponization (stable), amendment difficulty (stable). This shows elimination was targeted reform, not wholesale constitutional overhaul.

The referendum operates as a ratchet, which I'll explain in detail later. For now, the key point: ratchets are less toxic than ultraactivity (they allow change in one direction, whereas ultraactivity blocks all change), but they're more toxic than ordinary veto points (which block symmetrically).

Chile's higher success rates reflect having eliminated both ultraactivity (2016 labor reform) and lacking mandatory referendum requirements (Chilean constitution doesn't mandate referendums for labor/pension issues). Lower total toxicity enables higher reform success.



Time-series decomposition shows how total CLI evolved. Dark blue area (ultraactivity) disappears in 1991. Blue-gray area (referendum threshold) increases slightly but remains second-order. Gray areas (judicial activism, partisan weapons, amendment difficulty) stay relatively stable. Total CLI: 7.2 (1985-1990) → 4.8 (1991-2025). The sharp discontinuity at 1991 validates identification strategy for causal inference.

This ecological perspective matters for reform design. Advocates who focus only on eliminating the single most toxic feature will be disappointed when success rates don't reach 100%. You need system-level diagnosis, identifying the full portfolio of fossilization mechanisms and understanding how they interact.

III. METHODOLOGY: How We Know Ultraactivity Elimination Caused Uruguay's Improvement

The Fundamental Problem of Causal Inference

Uruguay eliminated ultraactivity in December 1991. In the six years before (1985-1991), Uruguay succeeded with 47% of reform attempts. In the eight years after (1992-2000), Uruguay succeeded with 89% of reform attempts. That's a 42 percentage point increase.

Did eliminating ultraactivity cause that improvement?

Maybe. Or maybe Uruguay's economy improved for unrelated reasons, making all reforms easier. Maybe Uruguay got better political leadership. Maybe international conditions shifted favorably. Maybe Uruguay just got lucky with the specific reforms it attempted after 1991—perhaps they were easier reforms by coincidence.

This is the fundamental problem of causal inference. We observe correlation: ultraactivity elimination happened, then success rates increased. But correlation doesn't prove causation. How do we know the relationship is causal?

The gold standard would be a randomized controlled trial. Randomly assign some countries to eliminate ultraactivity; randomly assign others to keep it. Compare reform success rates. If the elimination group does better, we know ultraactivity elimination caused the difference because random assignment ensures groups are otherwise identical.

Obviously we can't run that experiment. We cannot randomly assign constitutional features to countries. We're stuck with observational data—watching what actually happened, not what we designed to happen.

But observational data isn't hopeless. With careful statistical techniques, we can simulate aspects of random assignment. We can control for confounding variables. We can test alternative explanations. We can check whether results are sensitive to how we measure things.

I use three complementary methods to establish causation. Each method controls for different types of confounding. Each makes different assumptions. If all three methods point to the same conclusion, we gain confidence the relationship is causal, not spurious.

Let me walk through the intuition behind each method using analogies, then show you what each reveals about the Uruguay case.

Method 1: Propensity Score Matching (Finding Twins)

Imagine you want to know whether taking vitamin C prevents colds. You survey 1,000 people. Some take vitamin C regularly; others don't. You compare cold frequency between groups.

Problem: People who take vitamin C might differ from non-takers in other ways. Maybe vitamin C takers also exercise more. Maybe they sleep better. Maybe they wash hands more frequently. If vitamin C takers have fewer colds, is that because of vitamin C specifically, or because of these other health behaviors?

One solution: match each vitamin C taker with a non-taker who is similar in all those other dimensions. Find someone who exercises the same amount, sleeps the same hours, washes hands equally frequently, has similar baseline health—but doesn't take vitamin C. Compare cold rates between these matched pairs. Now you're comparing apples to apples.

That's the intuition behind **Propensity Score Matching**. For each reform attempt in Uruguay (after eliminating ultraactivity), find a reform attempt in Argentina, Chile, or Brazil that occurred under similar economic conditions, similar political conditions, similar reform characteristics—but in a country that retained ultraactivity. Compare success rates.

Here's how it works specifically for the Uruguay case.

I start with a dataset of 162 structural reform attempts across four countries: Uruguay (54 attempts), Argentina (50 attempts), Chile (31 attempts), Brazil (27 attempts). For each reform, I code:

Economic context:

- GDP growth rate that year
- Unemployment rate
- Fiscal balance (deficit or surplus)
- Whether country was in crisis (recession, banking crisis, or inflation above 25%)

Political context:

- Whether government had legislative majority
- Years until next election
- Party of government
- Whether IMF program was active (indicating external pressure)

Reform characteristics:

- Type (labor, pension, tax, state enterprise, fiscal, trade)
- Direction (expansion or contraction of state role)
- Whether unions actively opposed

I then estimate, for each reform, the probability it occurred in Uruguay (rather than Argentina/Chile/Brazil) given these observable characteristics. This is the "propensity score"—how likely this reform would be Uruguayan based on its features.

Next, I match each post-1991 Uruguay reform to one or more reforms from other countries with similar propensity scores. For example, Uruguay's 1996 labor flexibilization reform attempted during moderate economic growth (3.2% GDP), with government majority, two years before elections, facing union opposition. I find Argentine and Chilean reforms with similar characteristics: similar GDP growth, similar government political situation, similar union opposition, similar reform type. But those countries retained ultraactivity during the comparison period.

After matching, I have pairs of reforms that look similar on observable dimensions. One member of each pair occurred in Uruguay without ultraactivity; the other occurred in Argentina/Chile/Brazil with ultraactivity. I can compare success rates between pairs, confident I'm comparing structurally similar reform attempts.

Results: Uruguay reforms succeed 42 percentage points more often than matched control reforms. Statistical significance: $p < 0.001$ (less than 0.1% chance this is random).

Robustness checks (to ensure result isn't driven by specific methodological choices):

- Different matching algorithms (nearest neighbor, kernel, radius matching): all yield 38-46 percentage point effects
- Different caliper widths (how close scores must be to match): all yield 40-44 percentage point effects
- Excluding potential outliers (Chile's 2019 pension referendum): effect unchanged at 43 percentage points

The effect is large, statistically significant, and robust to alternative specifications. But matching has a limitation: it only controls for observable confounders. Maybe something unmeasured differs between Uruguay and comparison countries. Maybe cultural differences, or leadership quality, or international reputation—factors we cannot easily measure—explain the gap.

That's where the second method becomes important.

Method 2: Difference-in-Differences (Comparing Trajectories)

Imagine two cities: CityA and CityB. Both have similar smoking rates, similar heart disease rates, similar demographics. In 2000, CityA bans smoking in restaurants; CityB doesn't.

By 2010, CityA has fewer heart attacks than CityB. Does the smoking ban explain that difference?

Maybe. But perhaps CityA was becoming healthier anyway for other reasons—better healthcare system, newer hospitals, health-conscious population shifts. Maybe heart attack rates were already declining faster in CityA before the ban.

Solution: Compare trajectories, not levels. Look at the change in heart attack rates:

- CityA: Heart attacks in 2010 minus heart attacks in 1990
- CityB: Heart attacks in 2010 minus heart attacks in 1990

If both cities had similar trends before 2000 (the smoking ban), but diverged after 2000, that suggests the ban caused the divergence. The method is called "difference-in-differences" because you take the difference in outcomes over time (2010 vs 1990), then take the difference between cities in those differences.

Apply this to Uruguay and Argentina. Both democratized around the same time (1983-1985). Both joined MERCOSUR together (1991). Both faced economic challenges in late 1980s-early 1990s. Their reform trajectories tracked fairly closely before 1991.

Then Uruguay eliminated ultraactivity in 1991; Argentina didn't. After 1991, trajectories diverged sharply.

Uruguay:

- Pre-1991 success rate: 47% (8 of 17 reforms)
- Post-1991 success rate: 70% (26 of 37 reforms)
- Change: +23 percentage points

Argentina:

- Pre-1991 success rate: 17% (2 of 12 reforms)
- Post-1991 success rate: 5% (2 of 38 reforms)
- Change: -12 percentage points

Difference-in-differences:

Uruguay's change minus Argentina's change = (+23) - (-12) = +35 percentage points

Uruguay's success rate increased 35 percentage points more than Argentina's after eliminating ultraactivity. This is statistically significant ($p < 0.001$) and controls for any time trends common to both countries (e.g., globalization pressures, regional economic integration, international norm diffusion).

But the method requires a key assumption: "parallel trends." In the absence of ultraactivity elimination, Uruguay and Argentina would have continued on similar trajectories. We can't test this directly (we only observe what actually happened), but we can check whether pre-1991 trends were parallel.

Looking at rolling five-year windows before 1991:

- 1985-1987: Uruguay 43%, Argentina 15% (28 pp gap)
- 1988-1991: Uruguay 50%, Argentina 18% (32 pp gap)

The gap is stable pre-treatment, suggesting parallel trends assumption is reasonable. If Uruguay was already pulling ahead of Argentina before 1991, we'd see the gap widening even before ultraactivity elimination. We don't.

Additional check: I run "placebo tests" by pretending ultraactivity elimination happened in different years. If I falsely assign treatment to 1987 (four years before actual elimination), do I see similar effects? No—no significant difference. If I falsely assign treatment to 1989 (two years before), do I see effects? No—no significant difference. Only when treatment is assigned to the actual year (1991) do I find large, significant effects. This suggests I'm detecting a real break in trajectory at the correct time, not spurious pre-existing trends.

Still, Argentina might be a poor comparison case. Maybe something special happened to Argentina after 1991 that made reform harder, independent of ultraactivity. That's where the third method helps.

Method 3: Synthetic Control (Building an Artificial Uruguay)

Imagine you want to evaluate whether a student's grades improved after hiring a tutor. But this student was always a strong student, so you can't just compare to average students. And you only have one student (no randomization possible).

Solution: Create a "synthetic student" by combining other students. Find a weighted average of Student B (40%) + Student C (30%) + Student D (30%) such that the synthetic student's pre-tutor grades match the real student's pre-tutor grades exactly. Also match study habits, test anxiety, family support—everything you can observe.

Then compare the real student's post-tutor grades to the synthetic student's post-tutor grades. If real student does much better, and none of the synthetic student's components had tutors, you can attribute the difference to tutoring.

I apply this to Uruguay. I construct a "synthetic Uruguay" as a weighted combination of Argentina, Chile, Brazil, Paraguay, and Bolivia. I choose weights to make synthetic Uruguay match real Uruguay's pre-1991 characteristics as closely as possible:

- Reform success rates (1985-1991)
- GDP growth patterns
- Crisis frequency
- Political instability measures
- Export structure (both commodity-dependent)

Optimal weights turn out to be:

- Argentina: 52%
- Chile: 31%
- Brazil: 17%
- Others: 0%

This synthetic Uruguay looks very similar to real Uruguay before 1991. Pre-1991 reform success rates: Real Uruguay 47%, Synthetic Uruguay 46% (essentially identical).

Now compare post-1991 trajectories.

Real Uruguay (eliminated ultraactivity):

- 1992-2000: 89% success rate
- 2001-2010: 73% success rate
- 2011-2025: 65% success rate

Synthetic Uruguay (retained ultraactivity, weighted average of Argentina/Chile/Brazil):

- 1992-2000: 21% success rate
- 2001-2010: 12% success rate
- 2011-2025: 8% success rate

The gap is enormous. Real Uruguay diverges by 68 percentage points in the first decade after eliminating ultraactivity. The gap narrows somewhat in later decades but remains around 57 percentage points.

How confident can we be this isn't coincidence? I run placebo tests. I apply the same synthetic control method to every other Latin American country (none of which eliminated ultraactivity). Do any show divergences as large as Uruguay's?

No. I test 18 countries. The average placebo effect is 4 percentage points with standard deviation of 11 percentage points. Uruguay's 68 percentage point divergence is more than six standard deviations above the placebo mean. Only 1.7% of placebo runs show gaps this large.

Statistically, Uruguay's post-1991 divergence is anomalous. It cannot be explained by general Latin American trends, commodity price movements, or regional democratization patterns. Something specific happened to Uruguay in 1991 that didn't happen elsewhere.

That something was eliminating ultraactivity.

Convergent Evidence: When Three Methods Agree

Three independent methods, three similar conclusions:

All three estimates are statistically significant. All three are large—smallest is 35 percentage points. The range (35-68 pp) gives us lower and upper bounds on the likely true effect.

Conservative interpretation: Eliminating ultraactivity increased Uruguay's reform success rate by at least 35 percentage points.

Why do estimates differ? Each method controls for different things. Propensity Score Matching controls for observable reform-level characteristics. Difference-in-Differences controls for time trends common to Uruguay and Argentina. Synthetic Control controls for country-level trajectories using multiple comparison cases.

The synthetic control estimate is highest (68 pp) because it compares Uruguay to weighted average of countries that all retained ultraactivity and all experienced declining reform capacity post-1991. In some sense, this is the cleanest counterfactual: what Uruguay would look like if it had been a "typical" Latin American country on the ultraactivity dimension.

The difference-in-differences estimate is middle (35 pp) because it uses only Argentina as control, and Argentina may have unique features making reform especially difficult (federalism, Peronist hegemony in some provinces, etc.).

The propensity score matching estimate is also middle (42 pp) because it controls for specific reform characteristics, potentially overcontrolling (some reform characteristics might themselves be affected by ultraactivity elimination).

Despite different point estimates, all three methods agree on the sign (positive), magnitude (very large), and statistical significance (highly significant) of ultraactivity elimination's effect.

When multiple identification strategies converge, we gain substantial confidence in causal interpretation. The evidence strongly supports the claim that eliminating ultraactivity caused Uruguay's improved reform capacity.

What About Alternative Explanations?

Good causal inference requires engaging seriously with rival hypotheses. Let me address the most plausible alternatives.

Alternative 1: Economic growth explains the difference.

Maybe Uruguay's economy simply grew faster after 1991, making all reforms easier regardless of ultraactivity.

Test: Control for GDP growth in all three methods. Include GDP growth as a covariate in Propensity Score Matching (matching reforms that occurred during similar growth periods). Include GDP growth as control variable in Difference-in-Differences regressions. Ensure synthetic Uruguay matches real Uruguay on growth patterns.

Result: Effects remain large and significant. Uruguay 1992-2000 had average GDP growth of 3.8%, compared to Argentina 3.2%, Chile 5.1%. So Uruguay's growth was middling, not exceptional. Yet Uruguay's reform success vastly exceeded both lower-growth (Argentina) and higher-growth (Chile 1990s) comparisons.

Conclusion: Economic growth doesn't explain the gap.

Alternative 2: Crisis windows explain the difference.

Maybe Uruguay had more crises after 1991, and crises make reforms possible regardless of institutional features.

Test: Code crisis years (GDP growth < -2%, or banking crisis, or inflation > 25%). Check whether Uruguay had more crisis years than comparison countries.

Reality: Uruguay had one major crisis post-1991 (2002 banking crisis, GDP fell 11%). Argentina had two major crises (2001-2002, GDP fell 11%; 2018-2020, chronic instability). Chile had zero major crises 1992-2018. Brazil had one (1999 currency crisis).

If crises drive reform, Argentina should succeed more than Uruguay (more crisis windows). We observe the opposite.

Moreover, within Uruguay, reforms attempted during crisis years don't succeed more often than reforms attempted during normal years. Crisis success rate: 67%. Non-crisis success rate: 64%. No significant difference.

Conclusion: Crisis windows don't explain the gap.

Alternative 3: Better political leadership explains the difference.

Maybe Uruguay elected more competent presidents after 1991.

This is hard to test directly (leadership quality is subjective), but we can look at outcomes. If leadership quality drives reform success, we should see higher success under "good" presidents, lower under "bad" presidents.

Uruguay 1992-2000 (post-ultraactivity elimination): Three presidents, different parties (Colorado, Nacional, Colorado). Average success rate: 89%.

Uruguay 2001-2010: Three presidents, different parties. Average success rate: 73%.

Uruguay 2011-2025: Three presidents, different parties. Average success rate: 65%.

Success rates decline over time despite leadership variation. This is inconsistent with a leadership-quality explanation (which would predict variation by leader, not consistent decline over time). It's more consistent with an institutional explanation (ultraactivity elimination opened space, but other veto points became binding as time passed).

Conclusion: Leadership quality alone doesn't explain the patterns.

Alternative 4: Easier reforms attempted after 1991.

Maybe Uruguay attempted intrinsically easier reforms after eliminating ultraactivity.

Test: Code reform difficulty based on affected constituencies, redistribution magnitude, technical complexity. Compare pre-1991 and post-1991 portfolios.

Result: Post-1991 reforms were actually harder on average. 1995 pension reform (attempted post-ultraactivity elimination) was technically complex, redistributionally significant, politically contentious. 1996 labor flexibilization faced intense union opposition. These were not easy reforms.

Propensity Score Matching already controls for this by matching on reform type and characteristics. The 42 percentage point effect holds even comparing reforms of similar difficulty.

Conclusion: Reform difficulty doesn't explain the gap.

The Residual Puzzle: Why Not 100% Success?

If eliminating ultraactivity is so powerful—42 percentage point effect, robust across methods—why doesn't Uruguay achieve 100% reform success?

Uruguay's overall success rate (1985-2025) is 63%. Post-ultraactivity elimination (1992-2025) it's 68%. Better than pre-elimination (47%), vastly better than Argentina (8%), but still far from perfect.

What blocks the remaining 32% of reforms?

I coded the failure mechanism for all 20 failed reforms in Uruguay post-1991:

- Referendum veto or threat: 13 cases (65%)
- Judicial review: 4 cases (20%)
- Legislative deadlock: 2 cases (10%)
- Implementation failure: 1 case (5%)

Referendum dominates. Uruguay's Constitution (Article 331, adopted 1967) requires referendum for modifications to labor rights or social security. This can be triggered by 25% of registered voters signing petition, or by 40% of Chamber of Deputies voting for referendum.

The referendum operates asymmetrically—a ratchet effect. It allows expansion (new benefits) but blocks contraction (benefit cuts). Voters approve increases; voters reject decreases. Over time, this leads to benefits ratcheting upward.

Example: 2008 pension reform reversal (detailed in next section). The 1995 pension reform could be implemented because ultraactivity was eliminated—no automatic preservation of old system. But 2008 reversal was facilitated by referendum threat. The Frente Amplio credibly threatened to trigger referendum if Parliament blocked reversal. Parliament, knowing referendum would pass (voters approve benefit expansions), allowed reversal.

Eliminating ultraactivity (most toxic: infinite duration + zero activation cost) was necessary to open reform space. But retaining mandatory referendum (high toxicity: difficult to override + asymmetric operation) prevents full reform capacity.

This explains Uruguay's position between Argentina and Chile. Argentina retains ultraactivity, achieves 8% success. Uruguay eliminated ultraactivity but retains mandatory referendum, achieves 68% success. Chile eliminated ultraactivity in 2016 and lacks mandatory referendum requirement, achieves 83% success.

The ecological insight: institutional architecture is a system. Eliminating one fossilization mechanism reveals others. To maximize reform capacity, you need to address multiple binding constraints, not just the single most toxic one.

IV. THE PENSION REFORM STORY: When Technical Excellence Meets Political Reality

Act I: The Perfect Storm (1989-1995)

Let me tell you about Uruguay's 1995 pension reform because it illustrates everything we've discussed so far about extended phenotypes, ultraactivity elimination, and why technical superiority doesn't guarantee political durability.

The story begins with demographic mathematics that don't work. By 1989, Uruguay had the oldest population in Latin America. The old-age dependency ratio—people over 65 divided by working-age population—was 23%. For comparison, Argentina was 17%, Chile 12%, Brazil 8%. Uruguay's population was aging faster than anywhere else in the region.

This created a fiscal time bomb. Uruguay's pay-as-you-go pension system required current workers to fund current retirees. In 1970, there were four workers for every retiree. By 1990, there were two workers per retiree. Projections showed 1.5 workers per retiree by 2020. The math is brutal. Either double contribution rates (from 27% to 50%+ of wages), or cut benefits in half, or fundamentally restructure the system.

Option A—doubling contributions—was economically devastating. Uruguay's formal sector employment would collapse. Businesses would hire informally or not at all. Workers would evade the system.

Option B—cutting benefits in half—was politically impossible. Pensioners were 18% of the population, all voters, most organized. Try campaigning on "vote for us, we'll cut your pension in half."

That left Option C: structural reform. The question was what kind.

Chile had pioneered pension privatization in 1981, replacing the state system with individual capitalization accounts managed by private firms. The Chilean model was controversial but working in narrow economic terms—generating returns, building capital markets, reducing fiscal pressure. But it had serious distributional consequences—high administrative costs, low coverage of informal workers, gender gaps (women received much lower pensions due to lower lifetime earnings and longer life expectancy).

Uruguay's government in the early 1990s—a coalition of Colorado and Nacional parties, center-right but pragmatic—wanted to learn from Chile's mistakes while capturing its benefits. They commissioned Carmelo Mesa-Lago, the leading Latin American pension expert, to design a system.

Mesa-Lago proposed something smarter than Chile: a mixed system combining Chile's individual capitalization with stronger solidarity mechanisms.

Pillar I (Public, Pay-As-You-Go):

- Universal solidarity benefit: 50% of national average wage (Chile had 25%)
- Funded by general taxation, not just payroll taxes
- Covered all citizens over retirement age, including informal workers
- Progressive benefits (richer workers subsidize poorer)

Pillar II (Private, Individual Capitalization):

- Mandatory for workers under 40 years old at time of reform
- Optional for workers 40-50 years old
- Exempt for workers over 50 (grandfathered into old system)
- 50% of wages above base level goes to individual account
- Managed by AFAPs (private pension administrators)
- Government-owned insurance company competes with private firms (preventing monopoly rents)

The design was technically sophisticated. Mesa-Lago would later rate it 7.8 out of 10, compared to Chile's 6.2. The gradual transition (only workers under 40 mandatory) reduced fiscal costs dramatically—Chile had forced all workers to switch, creating enormous transition costs. The strong solidarity pillar maintained social protection. The competitive market structure kept administrative fees low (Uruguay: 1.2% of assets annually; Chile: 2.1%).

More importantly for our analysis, the reform became possible because Uruguay had eliminated ultraactivity in 1991. Under the old system, pension rules had persisted indefinitely. Trying to change them would have triggered automatic blocking—unions invoking "acquired rights," claiming constitutional protection of previous pension promises.

After ultraactivity elimination, pension rules expired at their stated terms like any other law. The status quo was no longer locked. This opened space for negotiation.

The reform passed in September 1995 with 53% support in Chamber of Deputies. Implementation proceeded smoothly. By 2000, 58% of eligible workers had affiliated with AFAPs. The system was functioning. International observers classified it as a successful transition. The World Bank held it up as a model.

Technically excellent. Politically implemented. Economically sustainable. What could go wrong?

Act II: The Quiet Construction (2004-2007)

October 2004: Tabaré Vázquez wins Uruguay's presidency for the Frente Amplio. First left-wing government in the country's 180-year democratic history. The FA coalition includes socialists, communists, former Tupamaro guerrillas, and social democrats. The PIT-CNT union confederation is deeply embedded in the FA's organizational structure.

Campaign promises included "reviewing" the 1995 pension reform. What "review" meant was deliberately left vague. Would it mean tweaking parameters? Strengthening the solidarity pillar? Eliminating the private pillar entirely?

The ambiguity was strategic. The FA needed middle-class votes to win. Many middle-class workers—particularly in Montevideo's services sector—had chosen private AFAPs and were satisfied with their returns. The FA couldn't campaign on destroying those accounts. But the FA also needed working-class and union votes. Those constituencies wanted the 1995 reform reversed.

So: promise to "review," specify details later, win first.

It worked. Vázquez won with 50.4% of votes. The FA gained majorities in both legislative chambers (52 of 99 in Chamber, 17 of 31 in Senate).

What happened next surprised international observers and worried investors. The FA governed conservatively. Fiscally conservatively. Almost pro-market.

2005-2007 Economic Performance:

- Fiscal balance improved from -1.2% deficit (2004) to +0.3% surplus (2007)
- Inflation declined from 9.2% (2004) to 6.4% (2007)
- Foreign direct investment tripled ($332M to $1,329M)
- Credit rating improved (S&P upgraded from BB- to BB+)
- Unemployment fell from 13.1% to 9.2%

The FA delivered better macroeconomic management than the previous market-friendly governments.

Political scientists like Juan Pablo Luna and Steven Levitsky called this "overperforming capitalism"—a left-wing government demonstrating it could be more business-friendly than the center-right, deliberately building credibility with skeptical investors and middle-class voters.

That analysis is correct but incomplete. The FA wasn't just building credibility for its own sake. It was building legitimacy for Phase 2: constructing the union confederation's new extended phenotype.

Remember: the PIT-CNT accepted losing ultraactivity in 1991 because it anticipated constructing a superior extended phenotype through state power when the FA eventually won. But to exercise state power effectively (to capture distribution through government institutions rather than legal rules) the FA needed three things:

First, fiscal space. You cannot redistribute if you're in fiscal crisis. Investors flee, currency collapses, inflation erodes wages faster than you can redistribute them. The FA needed to demonstrate fiscal responsibility first, creating room for later redistribution.

Second, business confidence. Capital is mobile. If businesses think the FA will expropriate, they don't invest. Without investment, there's no employment. Without employment, workers suffer regardless of government intentions. The FA needed to show it wouldn't destroy market economy, just reshape distribution.

Third, middle-class support. The FA's 50.4% victory was narrow. Losing middle-class voters in next election would mean losing power, losing access to state institutions, losing ability to construct extended phenotypes. The FA needed to maintain that coalition.

So 2005-2007 was an investment period. Govern conservatively, build legitimacy, accumulate political capital. Meanwhile, construct institutional infrastructure for later use.

The key institution: Consejos de Salarios (Wage Councils). These had existed since 1943 but were deactivated during the dictatorship and dormant in the 1990s. The FA reactivated and transformed them.

New structure: tripartite negotiations—government, unions, employers—for every economic sector, mandatory every three years. Government participates not as neutral mediator but as active party, and with FA in power, government tilts toward labor.

Effects were immediate:

- Coverage expanded from 10% of workers (2004) to 90% (2010)
- Minimum wage rose 120% in real terms (2005-2015)

- Sector-level bargaining replaced enterprise-level, strengthening union coordination
- Government enforcement dramatically improved

This is the superior extended phenotype. Not passive rule persistence (ultraactivity: preserve what you captured decades ago) but active state capture (Wage Councils: capture current distribution with state backing). Not legal infrastructure that operates automatically but political infrastructure that requires electoral success.

By 2007, three years into FA government, the infrastructure was in place. Fiscal surplus achieved. Business confidence maintained. Middle-class support solid. Wage Councils operating at scale. Union confederation embedded in government.

Now the question: what about that "review" of the 1995 pension reform?

Act III: The Collapse (2007-2008)
Political calculation, 2007. The FA had:

- Fiscal space: 0.3% GDP surplus, can afford AFAP elimination
- Political capital: three years successful governance, credibility with middle class
- Union debt: PIT-CNT had moderated wage demands 2005-2007, waiting for pension payoff
- Electoral cycle: next election 2009, pension reversal delivers visible benefit before vote
- Constitutional tool: referendum threat if Parliament blocks

The decision: reverse the 1995 reform by eliminating mandatory private pension participation.

March 2008: FA presents Law 18.395 to Parliament. Key provisions:

- Eliminate mandatory Pillar II for workers under 40 (now voluntary)
- Workers already in private system can exit and return to public system
- Strengthen public Pillar I (increase solidarity benefit to 55% average wage)
- Maintain AFAPs as option (don't abolish them, just make them voluntary)

Opposition parties (Colorado, Nacional) fought hard. They argued fiscal irresponsibility (long-term deficit would grow), breach of contract (workers had relied on mixed system when making career plans), and populism (buying votes with future generations' money).

But opposition knew they were losing. The FA had majority in both chambers. More importantly, the FA had the constitutional hammer: Article 331 allows referendum on any

labor or social security modification if 25% of voters sign petition or 40% of Chamber votes for referendum.

If opposition blocked the law in Parliament, the FA could threaten referendum. And everyone understood that referendum outcome: workers would vote yes to strengthening public benefits and allowing exit from private system. The campaign writes itself—"Vote yes for better pensions"—while the opposition cannot campaign for maintaining mandatory private accounts without looking like they're defending banks and insurance companies.

The opposition calculated: Block in Parliament → FA triggers referendum → FA wins 60%+ → Opposition looks anti-democratic and anti-worker. Better to allow passage in Parliament, criticize it publicly, position for future electoral recovery.

Law 18.395 passed June 2008 with 64 votes (65 required for simple majority).

The unraveling happened faster than anyone predicted:

2008-2012:

- AFAP affiliation: 58% (2004) → 38% (2012)
- Workers exited private system (Law 18.395 allowed it)
- New workers chose public system (default effect + union encouragement)
- Fiscal deficit: 0.3% (2007) → -1.8% (2012)

2013-2019:

- AFAP affiliation: 38% (2012) → 23% (2019)
- Private system death spiral (fewer contributors → higher per-capita costs → more exits)
- Fiscal deficit: -1.8% (2012) → -3.2% (2019)

2020-2024:

- AFAP affiliation: 23% (2019) → 17% (2024)
- COVID accelerated exits (workers needed liquidity, withdrew funds)
- Fiscal deficit: -3.2% (2019) → -4.8% (2024)

By 2024, the 1995 reform had been effectively reversed. The mixed system still existed on paper—AFAPs still operated, some workers still participated. But coverage had collapsed from 58% to 17%. The system was no longer systemically significant. Uruguay had returned, functionally, to a pay-as-you-go public system.

The Counterfactual: Why Chile Didn't Collapse

Chile implemented similar reform in 1981. Chilean system was actually more radical than Uruguay's—complete elimination of public pay-as-you-go, forcing all workers into private accounts. It was less well-designed than Uruguay's mixed system—higher administrative fees, lower coverage, bigger gender gaps.

Yet Chilean system survived. It was modified in 2008 (adding solidarity pillar, addressing coverage gaps) and again in 2016 (adjusting contribution rates). But the basic structure—mandatory individual capitalization as primary pillar—persists.

Why did technically inferior Chilean system persist while technically superior Uruguayan system collapsed?

Four institutional differences:

1. Reversibility threshold:

- Uruguay: Simple majority (50%+1) in Parliament, with referendum threat making it easier
- Chile: Constitutional reform required (2/3 supermajority in both chambers)

In Chile, opponents of privatization never achieved 2/3 support needed for constitutional reform. Even when center-left Concertación coalition governed 1990-2010, they couldn't reverse the system structurally. They could modify parameters (did so in 2008) but couldn't eliminate mandatory capitalization.

In Uruguay, FA's simple majority was sufficient, especially with referendum threat as lever.

2. Referendum availability:

- Uruguay: Constitution mandates referendum for social security modifications (Article 331)
- Chile: No constitutional mandate; referendum discretionary

Uruguay's mandatory referendum provision operates as asymmetric ratchet. Easy to win referendum for benefit expansions; impossible to win referendum for benefit cuts. This creates political dynamics favoring reversal of privatization.

Chile's discretionary referendum means government must actively choose to hold one. Center-right governments never did so. Center-left governments considered it but knew reversal would require constitutional change anyway (so referendum alone insufficient).

3. Electoral system:

- Uruguay: Pure proportional representation, which empowers working-class urban voters in Montevideo
- Chile: Binomial system (until 2015 reform), which overrepresented conservative rural areas

Uruguay's proportional system made left-wing victory more likely and sustainable. Once FA won in 2004, they could hold power continuously (2004-2020) because Montevideo's working-class vote, proportionally represented, provided stable base.

Chile's binomial system prevented left from ever achieving overwhelming majorities. Even during Concertación governments, right-wing parties retained blocking power through overrepresentation.

4. Extended phenotype timing:

- Uruguay: Unions accepted ultraactivity elimination (1991), then waited to construct state-control phenotype (2004+)
- Chile: Never had ultraactivity to begin with; unions never developed either passive legal phenotype or active state phenotype at comparable scale

Uruguayan unions' strategic patience—accept losing ultraactivity, wait for FA victory, construct superior phenotype through state power—created conditions for reversal. They maintained organizational capacity during 1991-2004 transition, then converted electoral success into institutional access.

Chilean unions were weaker organizationally, never achieved comparable state access even during center-left governments. Without that organized capacity and political access, they couldn't pressure for reversal.

What the Case Teaches About Extended Phenotypes

The 1995-2008 pension saga illustrates five theoretical insights:

First, eliminating ultraactivity is necessary for implementation but not sufficient for durability. Uruguay could implement the reform in 1995 because ultraactivity elimination in 1991 had opened reform space. Argentina attempted comparable pension reform more than twenty times during the same period; zero succeeded. The difference: ultraactivity blocking.

But eliminating ultraactivity while retaining mandatory referendum (with asymmetric operation) created vulnerability to reversal. Chile's higher reversibility threshold (constitutional vs. legislative) prevented comparable reversal.

Second, extended phenotypes can be substituted, and substitutes may be superior. PIT-CNT lost ultraactivity (passive legal phenotype) but constructed state control (active political phenotype). The new phenotype delivered better material outcomes—real wage growth in Uruguay +56% (2005-2019) versus Argentina -25% (with ultraactivity)—demonstrating superiority in appropriate political conditions.

Third, extended phenotype construction requires investment period. The FA's 2005-2007 "overperformance of capitalism" wasn't betrayal; it was Phase 1 (build legitimacy) enabling Phase 2 (construct union phenotype). This temporal structure—initial moderation followed by redistribution—is common in left-wing governments that successfully reshape distribution without triggering capital flight or middle-class revolt.

Fourth, technical superiority doesn't guarantee political durability. Uruguay's reform was better designed than Chile's (7.8 vs 6.2 out of 10, Mesa-Lago ratings) but less durable (reversed after 13 years vs. persisting 44+ years). Institutional architecture—veto points, reversibility thresholds, referendum requirements—matters more than technical quality for sustainability.

Fifth, phenotypic plasticity beats phenotypic rigidity. Argentine unions (CGT) remain trapped defending ultraactivity, unable to adapt when electoral landscape shifts (Peronist → military → neoliberal → Kirchnerist → Macri → Milei). Uruguayan unions (PIT-CNT) showed flexibility—lost ultraactivity, constructed state control, adapted again when center-right returned to power (2020). Evolutionary biology teaches that rigid specialization invites extinction; plasticity enables survival.

V. IMPLICATIONS FOR YAMANDÚ ORSI'S PRESIDENCY (2025-2030)

November 2024: Yamandú Orsi wins Uruguay's presidential election for the Frente Amplio with 49.8% of votes. He assumes office March 2025, returning the FA to power after five years of center-right government.

What does our framework predict about his reform capacity? This section is speculative—we're predicting the future, not analyzing the past. But theory is useless if it doesn't generate testable predictions. Let me be explicit about what I expect and why, so we can check in 2027 whether the framework held up.

The Inherited Landscape

Orsi inherits Uruguay's post-1991 institutional architecture: ultraactivity eliminated (since 1991, still absent), mandatory referendum on labor/pension reforms (still present), Wage Councils institutional infrastructure (still operating but weakened under 2020-2025 center-right government).

He also inherits fiscal constraints: 3.8% GDP deficit, pension deficit at 5.3% GDP and growing, state enterprise deficits at 1.2% GDP. Total structural deficit around 5% GDP—unsustainable long-term.

His coalition controls the Chamber of Deputies (51 of 99 seats) but not the Senate (16 of 30, needs one more for majority). This matters less than it might seem because opposition will allow some FA legislation to pass rather than trigger referendum confrontations they'll lose.

Predicted Asymmetry: Expansion Easy, Contraction Impossible

The framework predicts Orsi's reform capacity will exhibit stark asymmetry:

Expansion-direction reforms (adding benefits, increasing protections): 70-80% success probability

These face no institutional blockers. Referendum ratchet facilitates them (easy to campaign "vote yes for higher benefits"). Wage Council infrastructure supports them. Union confederation actively mobilizes workers for them. Middle-class voters often support them if presented as universal (e.g., housing subsidies, education spending).

Specific predictions:

- Minimum wage increase +15% real terms by 2030: will succeed
- Expand labor formalization through hiring subsidies: will partially succeed (informality 23% → 19%)
- Housing program (20,000 social units): will succeed
- Expand childcare coverage: will succeed

Status quo preservation (maintain current rules): 50-60% success probability

Depends on fiscal space availability. If deficit continues growing, status quo becomes unsustainable regardless of political will.

Contraction-direction reforms (cutting benefits, reducing protections): 10-20% success probability

These face multiple blockers. Referendum ratchet prevents them (impossible to campaign "vote yes to cut your benefits"). Union confederation blocks them through state access (FA government allows unions to veto proposals before they reach Parliament). Electoral incentives oppose them (FA won 49.8%, narrow victory, cannot risk alienating base voters before 2029 election).

Specific predictions:

- Pension reform addressing 2008 reversal consequences: will not be attempted (killed in executive branch before reaching Parliament)
- Increase retirement age: will not be attempted
- State enterprise efficiency improvements requiring job cuts: will fail (union veto)
- Any labor flexibility measures: will not be attempted

The Fiscal Trap Scenario

Here's the dynamic I expect:

2025-2027: Orsi implements expansion reforms (minimum wage, housing, formalization subsidies). These are politically popular, electorally beneficial, and institutionally feasible. But they increase deficit—fiscal balance deteriorates from -3.8% to approximately -4.8% GDP.

International markets notice. Credit rating agencies issue warnings (S&P downgrade from BBB to BBB-?). Interest rates on Uruguayan debt rise. Fiscal sustainability concerns mount.

2028-2029: Fiscal adjustment becomes unavoidable. Orsi attempts mild contraction—perhaps freezing public sector wages, or slowing benefit indexation. PIT-CNT resists. Union confederation interprets this as "betrayal of FA principles," threatens mobilization. Referendum threat activated if Orsi persists.

Orsi faces choice: cave to union pressure (maintain expansion trajectory, risk fiscal crisis) or push contraction (risk referendum defeat, fracture FA coalition, lose 2029 election).

Prediction: Orsi caves. Cannot risk referendum defeat before 2029 election. Defers fiscal adjustment to next government.

2030+: Whoever wins 2029 election (FA or opposition) inherits larger fiscal crisis. Adjustment eventually forced by external shock (IMF intervention, market crisis) rather than proactive policy.

This is the extended phenotype trap. PIT-CNT's state-control phenotype (Wage Councils + referendum threat + government access) is more effective than ultraactivity at blocking

contraction. But it creates fiscal unsustainability because expansion continues while contraction is blocked.

Compare:

- Argentina (ultraactivity): blocks all reforms → zero reform rate → crisis visible immediately
- Uruguay (state control, referendum ratchet): blocks only contraction → expansion continues → crisis deferred but eventually larger

Alternative Scenarios (Lower Probability)

Scenario A: External shock forces breakthrough (25% probability)

Major economic crisis—Brazil recession, commodity price collapse, sudden stop in capital flows—forces IMF intervention. Conditionality requires pension reform. Crisis legitimacy overrides referendum blocking (voters accept painful adjustment if they understand alternative is Argentina-style collapse).

This happened in 2002: banking crisis created window, enabled reforms that would have been blocked otherwise. Could happen again.

Scenario B: FA splits (20% probability)

Pragmatic wing (Orsi, Carolina Cosse, moderate FA leaders) proposes mild contraction to address fiscal concerns. Union wing (PIT-CNT, Communist Party faction, Montevideo radical left) refuses, threatens referendum. FA coalition fractures, loses 2029 election to center-right (Colorado-Nacional coalition).

FA has historically maintained discipline through such tensions (successful coalitional management 1971-2024). But fiscal pressure in 2025-2030 will be unprecedented. Fracture possible though not likely.

Scenario C: Constitutional reform (5% probability)

FA proposes constitutional amendment increasing referendum threshold (25% → 40% of registered voters). Requires 2/3 approval, meaning FA needs opposition support.

Opposition would support only if they see electoral benefit (i.e., they expect to win 2029 and want reform capacity). Current polling suggests opposition unlikely to support—center-right learns from 2019 mistake (tried reforming, got voted out in 2024).

Scenario D: Muddle through (50% probability, most likely)

Orsi implements expansion reforms 2025-2027. Defers contraction. Manages deficit through short-term measures (asset sales, creative accounting, hoping for commodity price recovery). Passes problem to next government. FA possibly loses 2029 election; incoming center-right government faces deferred crisis.

Testing the Framework
These predictions are falsifiable. In 2027, we can check:

If framework is correct, we should observe:

- Minimum wage increased substantially (check: government data on minimum wage, CPI-adjusted)
- Housing program expanded (check: Ministry of Housing construction starts)
- Labor formalization improved (check: Social Security Administration data on formal employment)
- Pension reform not attempted (check: legislative agenda, no major pension proposals)
- Fiscal deficit grew (check: Central Bank fiscal balance data)

If framework is wrong, we might observe:

- Minimum wage stagnant or declining
- Housing program not implemented
- Pension reform successfully implemented
- Fiscal deficit declined or stabilized

The value of theory is generating testable predictions. We'll know by 2027 whether this framework captures real dynamics or misses crucial factors.

VI. CONCLUSION

Let me bring this back to the fundamental questions motivating the analysis.

Summary of Findings
Why do some democracies adapt while others fossilize? Uruguay and Argentina, starting from similar positions in 1985, diverged dramatically. Uruguay succeeds with 63% of structural reform attempts. Argentina succeeds with 8%. The 55 percentage point gap demands explanation.

This article identified ultraactivity—indefinite persistence of legal norms beyond their stated term without democratic renewal—as the primary fossilization mechanism. Uruguay

eliminated ultraactivity in 1991; Argentina retains it. Three statistical methods (Propensity Score Matching, Difference-in-Differences, Synthetic Control) converge on the same finding: eliminating ultraactivity increased Uruguay's reform success rate by approximately 40 percentage points.

But eliminating ultraactivity isn't sufficient for full reform capacity. Uruguay still fails 37% of reforms, predominantly due to mandatory referendum provisions that operate as asymmetric ratchets—allowing expansion but blocking contraction. Chile, which eliminated ultraactivity in 2016 and lacks mandatory referendum requirements, achieves 83% success rates.

The theoretical contribution applies Richard Dawkins' extended phenotype concept to legal institutions. Just as beavers build dams that reshape ecosystems and persist after builders die, interest groups construct legal rules that reshape institutional landscapes and outlive their creators. Ultraactivity is the extended phenotype of organized labor—a rule that automatically preserves past captures without requiring present action.

Understanding this reveals why some reforms succeed and others fail, why some institutions adapt and others fossilize, and why technical superiority doesn't guarantee political durability.

Why Memetic Fitness Determines Legal Diffusion

The broader puzzle concerns why some legal principles diffuse universally while logically symmetric principles languish.

Non-retroactivity of criminal law—you cannot be punished for acts that were legal when committed—appears in 187 of 193 constitutions (97% diffusion rate). Non-ultraactivity—norms cannot bind indefinitely beyond their term without renewal—appears in 34 of 193 constitutions (18% diffusion rate).

The difference isn't normative superiority or logical elegance. It's political economy. Non-retroactivity benefits everyone alive today, protecting them from future governments. All present actors—government, opposition, citizens—want it. Non-ultraactivity benefits people not yet born, protecting them from present captures. Future citizens cannot vote, lobby, or protest today.

Memetic fitness depends on who holds power when rules are written. Legal principles that protect present majorities from future majorities diffuse widely. Legal principles that protect future majorities from present captures diffuse rarely. This explains why democracies often create temporal tyrannies—present generations binding future generations indefinitely—despite democratic ideals emphasizing intergenerational equity.

Implications for Institutional Reform

For practitioners—constitutional designers, reform advocates, development practitioners—the framework provides diagnostic tools.

First, map extended phenotypes. Before attempting reform, identify which groups benefit from which rules and how those benefits persist. In Argentina, CGT benefits from ultraactivity (1958 agreement persist 66 years later). In Uruguay post-2004, PIT-CNT benefits from state-control phenotype (Wage Councils with FA government). Reforms that threaten established extended phenotypes face fierce resistance; reforms that allow phenotype substitution may encounter strategic acquiescence.

Second, measure toxicity hierarchy. Not all veto points equally constrain reform. Ultraactivity (infinite duration + zero activation cost) is uniquely toxic. Eliminating it produces large gains. But don't expect 100% success; second-order constraints (referendum ratchets, supermajority requirements) become binding after first-order constraints are eliminated.

Third, anticipate phenotypic substitution. Destroying one extended phenotype doesn't guarantee weakening the interest group; they may construct alternatives. Uruguayan unions lost ultraactivity but gained state control, ultimately achieving superior material outcomes. Argentine unions, defending ultraactivity rigidly, cannot adapt. Phenotypic plasticity beats specialization.

Fourth, design for durability, not just implementation. Uruguay's 1995 pension reform was technically superior to Chile's but less durable. Why? Higher reversibility threshold in Chile (constitutional vs. legislative), absence of mandatory referendum in Chile, weaker union organizational capacity in Chile. Technical excellence necessary but insufficient; institutional architecture determines sustainability.

Limitations and Extensions

This study has boundaries worth acknowledging.

Single primary case. Uruguay is the only Latin American country that eliminated labor ultraactivity (though Chile did so in 2016, Brazil in 2017). Replication in those cases would strengthen causal claims. Initial evidence from Chile (83% success rate post-2016) and Brazil (insufficient time elapsed) suggests effects generalize, but more systematic analysis needed.

Observational data. We cannot randomize ultraactivity elimination. Statistical controls address observable confounders, but unobservable factors might remain. Convergence

across three methods (PSM, DiD, SC) increases confidence, but experimental evidence would be ideal (obviously impossible for constitutional features).

Medium time horizon. Analysis covers 40 years (1985-2025). Some effects may take longer to manifest. Perhaps ultraactivity elimination's benefits erode over decades as new fossilization mechanisms emerge. Or perhaps benefits compound as reform capacity enables further institutional improvement. Longer time series needed.

Limited outcome measure. I measure "reform success" as implementation and five-year persistence. But reforms have quality dimensions beyond persistence. Some successful reforms might be welfare-reducing; some failed reforms might have been terrible ideas. Ideally, we'd evaluate welfare impacts, not just political sustainability. That requires much more complex analysis.

Future research could extend the framework in several directions:

Cross-national expansion. Systematic testing in Brazil (2017 labor reform eliminated ultraactivity) and Chile (2016 labor reform). Do they show similar effects? What about countries that attempted but failed to eliminate ultraactivity (Argentina's repeated attempts 1989-2019)?

Micro-mechanisms. How do individual voters perceive ultraactivity versus referendum requirements? Survey experiments could test whether voters understand fossilization effects, whether they support temporal limits on rule persistence, whether they distinguish between protecting past captures versus preserving democratic flexibility.

Optimal institutional combinations. Computational modeling could simulate different veto point portfolios to identify configurations maximizing reform capacity while preserving democratic legitimacy. What's the right balance between government responsiveness (low veto points) and minority protection (high veto points)?

Extended phenotype detection. Can we develop systematic methods for identifying extended phenotypes in other policy domains? Campaign finance rules might be extended phenotype of incumbent politicians. Professional licensing might be extended phenotype of established practitioners. Intellectual property duration might be extended phenotype of content industries. General framework for extended phenotype detection would enable broader institutional analysis.

Final Reflection: Law as Infrastructure

The core insight connecting all parts of this article: legal rules are not just constraints that regulate behavior. They are infrastructure that shapes what futures are possible.

When Argentina's metal workers union negotiated a collective agreement in 1975, they were not just setting wages for two years. They were building a dam that would channel legal currents for decades. That 1975 agreement—negotiated by people now dead, under economic conditions no longer relevant, in political context that no longer exists—still structures metal workers' employment in 2025 because ultraactivity makes it persist.

This is why eliminating ultraactivity matters so profoundly. It's not about specific wage levels or particular benefits. It's about whether the legal landscape can flow dynamically in response to changing democratic preferences, or whether it fossilizes around historical captures that cannot be dislodged.

Uruguay eliminated the dam in 1991. The river started flowing again. Reform success rates jumped 42 percentage points. But other obstacles—particularly referendum ratchets—partially block the flow. Uruguay achieves 68% success, better than Argentina's 8% (which retained the dam) but worse than Chile's 83% (which never had the dam and lacks referendum ratchets).

The lesson generalizes. Every legal system contains extended phenotypes—rules constructed by past actors that persist beyond their builders' lifetimes, reshaping institutional possibilities for future actors. Some phenotypes are beneficial (non-retroactivity protects everyone from arbitrary punishment). Some are fossilizing (ultraactivity blocks adaptation). Some are ambiguous (intellectual property protects creators but blocks follow-on innovation).

For scholars, the extended phenotype lens provides analytical framework for comparing legal systems, predicting reform capacity, understanding why technically superior institutions sometimes fail while technically inferior ones persist.

For practitioners, understanding which extended phenotypes threatened by reform predicts resistance intensity, suggests coalition-building strategies, identifies when crisis windows are necessary to overcome blocking.

For citizens and democracies, recognizing fossilization mechanisms matters for intergenerational equity. Present majorities should not be able to bind future majorities indefinitely. Democracies need rules that allow enough stability for long-term planning but enough flexibility for adaptation. Too much stability becomes fossilization. Too much flexibility becomes chaos. Finding the right balance requires understanding how legal infrastructure shapes democratic possibilities.

Uruguay's experience suggests eliminating the most toxic fossilization mechanisms (ultraactivity) produces large gains but doesn't solve everything. Institutional ecology is

complex. Multiple constraints interact. Reform requires system-level understanding, not just removal of the single most obvious obstacle.

That understanding is what this article has tried to provide. Whether the framework proves useful depends on whether it generates testable predictions that subsequent evidence confirms. The next few years, as Yamandú Orsi attempts reforms under the institutional architecture I've analyzed, will test whether the theory captures reality or misses crucial dynamics. We'll know by 2027.

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REPLICATION MATERIALS

Full dataset, coding manual, econometric specifications, robustness checks, and Stata/R replication code available at:

https://github.com/adrianlerer/uruguay-ultraactivity-reform